A Fisher-Information Reduction of the Cyclic Phase-Type Moment Hierarchy

1 The reduction

Let T be the absorption time of a phase-type chain with s transient states, and write

$$m_j = \mathbb{E}T^j.$$

For $r \geq 0$, let $T^{[r]}$ have the r-fold size-biased density

$$g_r(t) = \frac{t^r f(t)}{m_r}, \qquad t > 0.$$

Then

$$\mathbb{E}T^{[r]} = \frac{m_{r+1}}{m_r}, \qquad \mathrm{CV}^2(T^{[r]}) = \frac{m_r m_{r+2}}{m_{r+1}^2} - 1.$$

Consequently the desired hierarchy at $k = r + 1$,

$$m_{k-1}m_{k+1} \geq \frac{s+k}{s+k-1} m_k^2,$$

is equivalent to

$$\mathrm{CV}^2(T^{[r]}) \geq \frac{1}{s+r}. \tag{1}$$

The following analytic condition is sufficient for all the inequalities (1) simultaneously.

Proposition 1 (Log-convex density quotient implies the hierarchy)**.** *Suppose*

$$q(t) := \frac{f(t)}{t^{s-1}} \tag{2}$$

is log-convex on $(0, \infty)$*. Then* (1) *holds for every* $r \geq 0$*.*

Proof. Put $d = s + r$. Apart from its normalizing constant, the density of $T^{[r]}$ is

$$g(t) = t^{d-1}q(t).$$

Let

$$\ell(t) = \frac{q'(t)}{q(t)}.$$

The score for the log-scale family $g_a(t) = e^a g(e^a t)$ at $a = 0$ is

$$S(t) = 1 + t\frac{g'(t)}{g(t)} = d + t\ell(t).$$

Integration by parts gives

$$\mathbb{E}S = 0, \qquad \mathbb{E}[t\ell(t)] = -d. \tag{3}$$

Log-convexity of q says

$$\ell'(t) \geq 0, \qquad \text{equivalently} \qquad \ell(t)^2 \leq \frac{q''(t)}{q(t)}.$$

Two integrations by parts therefore give

$$\mathbb{E}[(t\ell(t))^2] \leq \frac{\int_0^\infty t^{d+1} q''(t)\, dt}{\int_0^\infty t^{d-1} q(t)\, dt} = d(d+1).$$

It follows from (3) that the scale Fisher information is

$$I = \mathbb{E}[S^2] = \mathbb{E}[(t\ell(t))^2] - d^2 \leq d.$$

Finally, another integration by parts yields

$$\mathbb{E}[(T^{[r]} - \mathbb{E}T^{[r]})S(T^{[r]})] = -\mathbb{E}T^{[r]}.$$

Cauchy–Schwarz now gives

$$\mathrm{Var}(T^{[r]})I \geq (\mathbb{E}T^{[r]})^2.$$

Since $I \leq d = s + r$, this is (1). □

All boundary terms in this calculation vanish for a phase-type density, including when the representation permits immediate absorption or the density has a lower-than-$s-1$ order zero at the origin.

2 Uniformization and the one missing matrix lemma

Write a phase-type density as

$$f(t) = \alpha e^{Qt} b, \qquad b = -Q\mathbf{1}.$$

Choose a uniformization rate $\lambda \geq \max_i(-Q_{ii})$, and put

$$P = I + \lambda^{-1} Q \geq 0, \qquad A = \lambda P \geq 0.$$

Then

$$f(t) = e^{-\lambda t} \alpha e^{tA} b.$$

Multiplication by $e^{-\lambda t}$ does not change the second derivative of the logarithm. Thus Proposition 1 would follow from the next standalone statement.

Conjecture 2 (Dimension-corrected matrix-exponential log-convexity)**.** *Let A be an entrywise nonnegative $s \times s$ matrix and let $u, v \geq 0$. If*

$$F(t) = u^T e^{tA} v$$

is not identically zero, then

$$t^{1-s} F(t) \quad \textit{is log-convex on } (0, \infty). \tag{4}$$

Equivalently,

$$t^2 \big(F(t)F''(t) - F'(t)^2\big) + (s-1)F(t)^2 \geq 0. \tag{5}$$

There is an equivalent local differential formulation which removes the two endpoint vectors. It appears well suited to an induction on the directed graph.

Proposition 3 (Growth-rate formulation)**.** *Let $h(t) = e^{tA}v$, initially with $v > 0$, and put*

$$r_i(t) = \frac{h_i'(t)}{h_i(t)}, \qquad c_{ij}(t) = A_{ij}\frac{h_j(t)}{h_i(t)}.$$

Then

$$r_i'(t) = \sum_j c_{ij}(t)\big(r_j(t) - r_i(t)\big). \tag{6}$$

Conjecture 2 is equivalent to the componentwise estimates

$$r_i'(t) \geq -\frac{s-1}{t^2} \qquad (1 \leq i \leq s). \tag{7}$$

Proof. Differentiating $r_i = h_i'/h_i$, using $h' = Ah$, gives

$$r_i' = \frac{\sum_j A_{ij}h_j r_j}{h_i} - r_i^2 = \sum_j c_{ij}(r_j - r_i),$$

because $\sum_j c_{ij} = r_i$. Taking $u = e_i$ in the matrix conjecture shows that (7) is necessary. Conversely, for arbitrary $u \geq 0$, set

$$\pi_i(t) = \frac{u_i h_i(t)}{u^T h(t)}.$$

A direct differentiation, or the usual log-sum identity, gives

$$(\log(u^T h))'' = \sum_i \pi_i r_i' + \sum_i \pi_i \left(r_i - \sum_j \pi_j r_j \right)^2.$$

Thus (7) implies $(\log(u^T h))'' \geq -(s-1)/t^2$. Vectors with zero entries follow by approximation. □

The coefficients c_{ij} in (6) are nonnegative and the matrix (c_{ij}) is diagonally similar to A. Moreover, along every directed cycle their product is independent of t, since the factors h_j/h_i telescope. Hence a negative r_i' can only be produced by flow from i toward vertices with smaller growth rate, while every closed cycle has zero total logarithmic change. The remaining problem is to turn this finite descending-chain structure into the sharp loss $(s-1)/t^2$.

Theorem 4 (Conditional all-k cyclic theorem)**.** *Conjecture 2 implies the phase-type moment hierarchy for every number of states s and every $k \geq 1$, including cyclic chains.*

Proof. Apply Conjecture 2 with $u^T = \alpha$ and $v = b$. Uniformization shows that the quotient $f(t)/t^{s-1}$ is log-convex. Proposition 1 then proves all the adjacent-moment inequalities. □

This isolates the cyclic problem in a form that no longer mentions absorption times, size biasing, or moments. In fact, if the moment hierarchy is required for every exponential tilt of every phase-type law, the matrix statement is also necessary. We record the precise converse next.

Lemma 5 (Laplace concentration)**.** *Let H be positive and C^4 near $t_0 > 0$, with enough exponential decay that the following integrals exist. Put*

$$J_{r,j} = \int_0^\infty t^{r+j} e^{-rt/t_0} H(t)\, dt \qquad (j = 0, 1, 2).$$

Then, as $r \to \infty$,

$$\frac{J_{r,0} J_{r,2}}{J_{r,1}^2} - 1 = \frac{1}{r} + \frac{t_0^2 (\log H)''(t_0) - 1}{r^2} + o(r^{-2}). \tag{8}$$

Proof. The probability measure proportional to $t^r e^{-rt/t_0} H(t)\, dt$ is a smooth tilt of a Gamma$(r + 1, r/t_0)$ law. Taylor-expand $\log H(t)$ at t_0, use the centered Gamma moments through order four, and expand its squared coefficient of variation. The constant and linear Taylor terms only change normalization and scale; the quadratic term gives $t_0^2 (\log H)''(t_0)/r^2$. The unperturbed Gamma law contributes $1/(r + 1) = r^{-1} - r^{-2} + O(r^{-3})$, which proves (8). Standard Laplace remainder estimates make the expansion uniform on a fixed neighborhood of t_0. □

Proposition 6 (The matrix lemma is forced by the tilted hierarchy)**.** *Suppose the phase-type hierarchy holds, for a fixed state bound s, for every k and every s-state phase-type law. Then Conjecture 2 holds for every nonnegative $s \times s$ matrix A and $u, v \geq 0$.*

Proof. It is enough first to take $u, v > 0$; the general case follows by approximation. Choose $\lambda > \rho(A)$, put

$$d = (\lambda I - A)^{-1} v > 0, \qquad D = \operatorname{diag}(d),$$

and define

$$Q = D^{-1} A D - \lambda I, \qquad b = D^{-1} v.$$

Then Q is a phase-type subgenerator and $-Q\mathbf{1} = b$. After multiplying u by an irrelevant positive constant, take $\alpha = u^T D$ to be a probability vector. Its absorption density is

$$\alpha e^{tQ} b = e^{-\lambda t} u^T e^{tA} v.$$

For every $\theta > 0$, exponentially killing this chain at rate θ and conditioning on absorption through the original exit b gives another s-state phase-type law. Its density is proportional to

$$e^{-\theta t} \alpha e^{tQ} b = e^{-(\lambda+\theta)t} F(t), \qquad F(t) = u^T e^{tA} v.$$

Indeed this is the usual Doob transform with $h = (\theta I - Q)^{-1} b$.

Fix $t_0 > 0$. For large r, choose $\theta = r/t_0 - \lambda > 0$, and apply the hierarchy at $k = r + 1$. The squared coefficient of variation of the r-fold size-biased law is the left side of (8) with $H = F$, whereas the hierarchy says that it is at least $1/(s + r)$. Hence

$$\frac{1}{r} + \frac{t_0^2 (\log F)''(t_0) - 1}{r^2} + o(r^{-2}) \geq \frac{1}{r} - \frac{s}{r^2} + O(r^{-3}).$$

Multiplying by r^2 and letting $r \to \infty$ gives

$$t_0^2 (\log F)''(t_0) + (s - 1) \geq 0.$$

Since t_0 was arbitrary, this is (5). □

3 The complete two-state cyclic case

The matrix conjecture, and hence the full hierarchy, can be proved unconditionally when $s = 2$.

Proposition 7. *If A is a nonnegative 2×2 matrix and $u, v \geq 0$, then*

$$\frac{u^T e^{tA} v}{t}$$

is log-convex on $(0, \infty)$.

Proof. Write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \mu = \frac{a+d}{2}, \qquad \delta = \sqrt{\left(\frac{a-d}{2}\right)^2 + bc}.$$

When $\delta > 0$,

$$e^{tA} = e^{\mu t}\left(\cosh(\delta t) I + \frac{\sinh(\delta t)}{\delta}(A - \mu I)\right).$$

The off-diagonal entries divided by t are nonnegative constant multiples of

$$e^{\mu t}\frac{\sinh(\delta t)}{\delta t}.$$

This is log-convex because

$$\frac{d^2}{dt^2}\log\frac{\sinh(\delta t)}{\delta t} = \frac{1}{t^2} - \delta^2 \operatorname{csch}^2(\delta t) \geq 0.$$

For a diagonal entry, put

$$h = \frac{a-d}{2\delta}, \qquad |h| \leq 1.$$

Apart from the common factor $e^{\mu t}$, that entry is either

$$\cosh(\delta t) + h\sinh(\delta t) = \frac{1+h}{2}e^{\delta t} + \frac{1-h}{2}e^{-\delta t}$$

or the same expression with h replaced by $-h$. It is therefore a positive sum of exponentials and is log-convex. Division by t preserves log-convexity because $-\log t$ is convex. The case $\delta = 0$ follows by continuity. Finally, $u^T e^{tA} v/t$ is a positive sum of these four log-convex entries, and positive sums of log-convex functions are log-convex. □

Corollary 8 (All moments for two cyclic states)**.** *Every phase-type law with a representation of size at most two satisfies*

$$m_{k-1} m_{k+1} \geq \frac{k+2}{k+1} m_k^2 \qquad (k \geq 1).$$

Proof. Apply Proposition 7 after uniformization and then use Proposition 1. □

4 Closure under block-triangular composition

The two-state result extends immediately to chains made from one- and two-state strongly connected components. The key closure principle is useful more generally.

Lemma 9 (Dimension-additive convolution)**.** *Let $p, q \geq 1$. Suppose $f, g \geq 0$, and suppose*

$$t^{1-p} f(t) \quad \text{and} \quad t^{1-q} g(t)$$

are log-convex. Then, for

$$(f * g)(t) = \int_0^t f(t-x) g(x)\, dx,$$

the function $t^{1-p-q}(f * g)(t)$ is log-convex.

Proof. Write $f(t) = t^{p-1}\phi(t)$ and $g(t) = t^{q-1}\psi(t)$. The substitution $x = ty$ gives

$$\frac{(f * g)(t)}{t^{p+q-1}} = \int_0^1 (1-y)^{p-1} y^{q-1} \phi(t(1-y))\psi(ty)\, dy.$$

For fixed y, the integrand is log-convex as a function of t. Positive integrals of log-convex functions are log-convex by Hölder's inequality. □

Proposition 10 (Block-triangular closure)**.** *Suppose Conjecture 2 holds in dimensions p and q. Then it holds for every nonnegative block upper-triangular matrix*

$$A = \begin{pmatrix} B & C \\ 0 & E \end{pmatrix}, \qquad B \in \mathbb{R}^{p \times p}, \quad E \in \mathbb{R}^{q \times q}.$$

Proof. It is enough to prove the assertion entrywise, since a positive sum of log-convex functions is log-convex. Entries in a diagonal block satisfy the claim by the corresponding lower-dimensional conjecture, followed by multiplication by t^{-q} or t^{-p}. For the upper-right block, Duhamel's formula gives

$$(e^{tA})_{ij} = \sum_{a,b} C_{ab} \int_0^t (e^{(t-x)B})_{ia} (e^{xE})_{bj}\, dx.$$

Every summand has the required dimension-$p+q$ correction by Lemma 9; summing preserves log-convexity. □

Corollary 11 (Components of size at most two)**.** *If every strongly connected component of the directed graph of a nonnegative $s \times s$ matrix A has at most two vertices, then $t^{1-s} u^T e^{tA} v$ is log-convex for all $u, v \geq 0$. Consequently the full moment hierarchy holds for every phase-type chain whose transient strongly connected components have size at most two.*

Proof. Permute the strongly connected components into block upper-triangular order and repeatedly apply Proposition 10. Components of size one are immediate, and components of size two are covered by Proposition 7. The phase-type conclusion follows from Proposition 1. □

5 An exact reduction of the three-state problem

There is a useful way to pass from an arbitrary entry of a positive three-dimensional semigroup to an absorption density. This removes the two endpoint vectors and makes the order-three canonical representation directly applicable.

Lemma 12 (A matrix entry is a phase-type density)**.** *Let A be an irreducible nonnegative $s \times s$ matrix and fix i, j. There is a transient Markov generator T with s transient states, an exit vector $b = -T\mathbf{1}$ supported only at j, and constants $c, \lambda > 0$ such that*

$$(e^{tA})_{ij} = ce^{\lambda t} e_i^T e^{tT} b.$$

Thus the logarithmic curvatures of the two sides are equal.

Proof. Choose $\lambda > \rho(A)$ and put

$$h = (\lambda I - A)^{-1} e_j = \sum_{n \geq 0} \lambda^{-n-1} A^n e_j.$$

Irreducibility gives $h > 0$. With $H = \mathrm{diag}(h)$, set

$$T = H^{-1}(A - \lambda I)H.$$

Its off-diagonal entries are nonnegative, and

$$T\mathbf{1} = -H^{-1}e_j, \qquad b = H^{-1}e_j.$$

The killing state is reachable from every transient state, so T is transient. Finally,

$$e_i^T e^{tT} b = h_i^{-1} e^{-\lambda t} (e^{tA})_{ij}.$$

Take $c = h_i$. Multiplication by a positive constant and by an exponential does not change logarithmic curvature. □

We now use the He–Zhang unicyclic representation theorem: every phase-type law of order three has a distribution-equivalent Markovian representation with subgenerator

$$T_c = \begin{pmatrix} -x_1 & 0 & \eta x_1 \\ x_2 & -x_2 & 0 \\ 0 & x_3 & -x_3 \end{pmatrix}, \qquad x_1 \geq x_2 \geq x_3 > 0, \quad 0 \leq \eta < 1. \tag{9}$$

Absorption occurs only from state 1, at rate $(1-\eta)x_1$. The canonical initial distribution need not be deterministic, but this causes no loss: after division by t^2 its density is a positive sum of the three deterministic-start densities.

Put

$$D(z) = (z + x_1)(z + x_2)(z + x_3) - \eta x_1 x_2 x_3.$$

A direct resolvent calculation gives their Laplace transforms:

$$\begin{aligned} L_1(z) &= \frac{(1-\eta)x_1(z + x_2)(z + x_3)}{D(z)}, \\ L_2(z) &= \frac{(1-\eta)x_1 x_2(z + x_3)}{D(z)}, \\ L_3(z) &= \frac{(1-\eta)x_1 x_2 x_3}{D(z)}. \end{aligned} \tag{10}$$

Consequently the whole 3×3 matrix problem has been reduced to the following scalar statement.

Conjecture 13 (Canonical cubic lemma)**.** *For the parameters in* (9)*, let* f_r *be the inverse Laplace transform of* L_r *in* (10)*. Then*

$$t^{-2}f_r(t) \quad \textit{is log-convex on } (0,\infty) \qquad (r=1,2,3).$$

Proposition 14 (The cubic lemma is exactly sufficient in dimension three)**.** *Conjecture 13 implies Conjecture 2 for every nonnegative* 3×3 *matrix.*

Proof. It is enough to treat matrix entries. Reducible matrices are already covered by Proposition 10. For an irreducible matrix apply Lemma 12, then replace the resulting order-three phase-type law by its He–Zhang representation. Its density is a positive sum of f_1, f_2, f_3. Positive sums preserve log-convexity, and the constant and exponential factors in Lemma 12 are harmless. □

The remaining lemma can also be written as a geometric-renewal problem. Let

$$g_r(t) = x_r e^{-x_r t}, \qquad C = g_1 * g_2 * g_3, \qquad R = \sum_{n\ge 0} \eta^n C^{*n},$$

where C^{*0} is a unit mass at zero. Then

$$f_1 = (1-\eta)g_1 * R, \qquad f_2 = (1-\eta)g_2 * g_1 * R, \qquad f_3 = (1-\eta)C * R. \tag{11}$$

For $\eta = 0$ the assertion follows immediately from Lemma 9: these are convolutions of respectively one, two, and three exponential densities, with extra factors t^{-2}, t^{-1}, and 1. What remains is therefore precisely the effect of geometric feedback.

There is also a complete coefficientwise proof when the three rates are equal. This supplies a nontrivial cyclic subcase and identifies the sharp root-of-unity calculation behind the directed three-cycle.

Proposition 15 (Equal-rate canonical cycle)**.** *Conjecture 13 holds when* $x_1 = x_2 = x_3$*.*

Proof. Scale the common rate to one. Apart from a positive constant and the common factor e^{-t}, the three densities in (11) are

$$H_r(t) = \sum_{n\ge 0} \eta^n \frac{t^{3n+r}}{(3n+r)!}, \qquad r = 0,1,2.$$

The exponential factor has no effect on logarithmic curvature. In the coefficient of η^N in

$$t^2(H_r H_r'' - (H_r')^2) + 2H_r^2,$$

all terms have degree $L = 3N + 2r$. After multiplication by the positive factor $L!$, that coefficient is

$$\sum_{\substack{0\le p\le L \\ p\equiv r \pmod 3}} \binom{L}{p}\left(2 - \frac{L}{2} + \frac{(L-2p)^2}{2}\right).$$

Let $\varepsilon = (-1)^N$. A roots-of-unity filter, applied first to $(1+x)^L$ and then to

$$(L - 2x\partial_x)^2(1+x)^L = L(1+x)^{L-2}\big(L(1-x)^2 + 4x\big),$$

evaluates the last display as

$$\frac{2^{L+1} + \varepsilon(-3L^2 + 3L + 4)}{3}. \tag{12}$$

This is nonnegative. For odd N it is immediate. For even N, the only borderline small case is $L = 4$, where equality holds; for $L \geq 5$ the elementary inequality $2^{L+1} \geq 3L^2 - 3L - 4$ follows by induction. The remaining cases $L = 0, 2$ are positive by inspection. Hence every coefficient of the curvature expression is nonnegative for $0 \leq \eta < 1$.

The same calculation gives a useful strengthening for the middle start. For $r = 1$, replace the correction $2H_1^2$ by H_1^2, as required for log-convexity of $t^{-1}H_1$. The filtered coefficient becomes

$$\frac{2^L + (-1)^N(-3L^2 + 3L + 2)}{3}, \qquad L = 3N + 2. \tag{13}$$

For odd N it is positive. For even N, equality holds at $L = 2$, and $2^L \geq 3L^2 - 3L - 2$ for $L \geq 8$. Hence $t^{-1}H_1(t)$ is log-convex in the equal-rate case. □

At this stage of the direct canonical route, the unequal-rate version of Conjecture 13 is the only unproved step. Formula (11) suggests proving that unequal exponential rates can only increase the curvature in the equal-rate certificate, but a valid polarization or sum-of-squares argument for that assertion is not yet known.

For further work it is useful to remove the negative diagonal without losing the special graph. Multiplication by $e^{x_1 t}$ does not change logarithmic curvature. Put

$$B = T_c + x_1 I = \begin{pmatrix} 0 & 0 & c \\ x_2 & a & 0 \\ 0 & x_3 & b \end{pmatrix}, \qquad a = x_1 - x_2, \qquad b = x_1 - x_3, \qquad c = \eta x_1,$$

so that $0 \leq a \leq b$. If $h = e^{tB}e_1$, define

$$z = c\frac{h_3}{h_1}, \qquad p = x_2\frac{h_1}{h_2}, \qquad q = x_3\frac{h_2}{h_3}.$$

The three component growth rates are

$$r_1 = z, \qquad r_2 = a + p, \qquad r_3 = b + q,$$

and satisfy the closed cyclic system

$$z' = z(b + q - z), \qquad p' = p(z - a - p), \qquad q' = q(a + p - b - q), \qquad zpq = cx_2x_3. \tag{14}$$

Thus the remaining lemma is equivalent to the three elementary-looking inequalities

$$t^2 z(z - b - q) \leq 2, \qquad t^2 p(a + p - z) \leq 2, \qquad t^2 q(b + q - a - p) \leq 2.$$

The initial limits are respectively $0, 1, 2$. Extensive numerical tests find no violation; in fact the middle expression appears bounded by 1. This Riccati formulation uses only five nonnegative scalar parameters and the ordering $a \leq b$, and is currently the most compact form of the missing unequal-rate argument.

The stronger middle bound also removes the third inequality altogether. More precisely, it is enough to prove

$$t^2 z(z - b - q) \leq 2, \qquad t^2 p(a + p - z) \leq 1. \tag{15}$$

Indeed, the second inequality says exactly that $t^{-1}h_2(t)$ is log-convex. Since

$$\left(e^{-bt}h_3(t)\right)' = x_3 e^{-bt}h_2(t), \qquad h_3(0) = 0,$$

we have

$$e^{-bt}h_3(t) = x_3 \int_0^t e^{-bs} h_2(s)\, ds.$$

Apply Lemma 9 with $p = 2$, $q = 1$, and with the second factor equal to the constant function 1. It follows that $t^{-2}e^{-bt}h_3(t)$, and hence $t^{-2}h_3(t)$, is log-convex. Thus only the first inequality and the sharper middle inequality in (15) are independent targets.

Two of the starts can in fact be completed whenever the canonical cubic has only real roots. Let

$$P(s) = s(s-a)(s-b) - Q, \qquad Q = cx_2x_3,$$

and suppose its roots are $\lambda_1 \le \lambda_2 \le \lambda_3$. The Laplace transforms of the shifted second and third components are

$$\mathcal{L}h_2(s) = \frac{x_2(s-b)}{P(s)}, \qquad \mathcal{L}h_3(s) = \frac{x_2x_3}{P(s)}.$$

Since $P(b) = -Q < 0$ and $P(s) \to \infty$ as $s \to \infty$, one has $\lambda_1 \le b < \lambda_3$. With

$$\theta = \frac{b - \lambda_1}{\lambda_3 - \lambda_1} \in [0, 1],$$

partial fractions give

$$\frac{s-b}{P(s)} = \frac{\theta}{(s-\lambda_1)(s-\lambda_2)} + \frac{1-\theta}{(s-\lambda_2)(s-\lambda_3)}. \tag{16}$$

The inverse transform of each summand, divided by t, is a positive mixture of exponentials, since

$$\frac{e^{ut} - e^{vt}}{u - v} = t \int_0^1 e^{t((1-y)v + yu)}\, dy.$$

Consequently $t^{-1}h_2(t)$ is log-convex, proving the sharper middle inequality in (15). Likewise, the inverse transform of $1/P(s)$ is the second divided difference of the exponential:

$$h_3(t) = x_2x_3t^2 \int_{\substack{y_i \ge 0 \\ y_1+y_2+y_3=1}} e^{t(y_1\lambda_1 + y_2\lambda_2 + y_3\lambda_3)}\, d\sigma(y).$$

Thus $t^{-2}h_3(t)$ is log-convex directly. Repeated roots follow by continuity. In the real-spectrum region, therefore, only the first canonical start remains open; the genuinely cyclic complex-pair region still requires the first and middle bounds in (15).

The remaining real-spectrum first start also has a useful positive-mixture form. Since

$$\frac{(s-a)(s-b)}{P(s)} = \frac{1}{s} + \frac{Q}{sP(s)},$$

the Hermite–Genocchi formula gives

$$h_1(t) = 1 + w(t), \qquad w(t) = Qt^3M(t), \qquad M(t) = \int_{\Delta_3} e^{tX(y)}\, dy, \tag{17}$$

where

$$X(y) = y_1\lambda_1 + y_2\lambda_2 + y_3\lambda_3, \qquad \Delta_3 = \{y_i \ge 0 : y_1 + y_2 + y_3 \le 1\}.$$

Let ν_t be the exponential tilt of normalized Lebesgue measure on Δ_3 by e^{tX}, and put

$$\theta = \frac{w}{1+w}, \qquad L = 3 + t\mathbb{E}_{\nu_t}X, \qquad V = t^2 \operatorname{Var}_{\nu_t}(X).$$

The mean under the tilt is increasing in t and starts at $(\lambda_1+\lambda_2+\lambda_3)/4 = (a+b)/4 \geq 0$; hence $L \geq 3$. Using $t\,d/dt$ in (17) gives the exact identity

$$-t^2(\log h_1)'' = \theta\big(3 - V - (1-\theta)L^2\big). \tag{18}$$

Consequently the first-start target is already proved whenever

$$\theta \leq \frac{1+\sqrt{3}}{3}, \qquad \text{equivalently} \qquad w \leq 5+3\sqrt{3},$$

because $V \geq 0$ and $L \geq 3$ make the right side of (18) at most $\theta(9\theta-6) \leq 2$. In the complementary range, the whole real-root problem is the scalar tilted-simplex inequality

$$V + (1-\theta)L^2 \geq 3 - \frac{2}{\theta}. \tag{19}$$

Thus no signed partial-fraction coefficients remain in the real-root target.

The real-root region also has a compact polynomial chart, and on a substantial initial range the target admits an exact finite certificate. After scaling, write the three roots as

$$m-\rho, \qquad m+\rho, \qquad m+1.$$

The coefficient identities for P and the nonnegativity of a, b, Q show that all three roots are nonnegative and that the admissible set is exactly

$$0 \leq m \leq 1, \qquad \max\left\{0, \frac{3m^2+2m-1}{4}\right\} \leq \rho^2 \leq m^2. \tag{20}$$

Indeed, if the two smaller roots were negative, their sum with the Perron root and their second elementary symmetric function could not both be nonnegative. The discriminant of the quadratic with roots a, b gives the lower bound in (20). This region is the union of the two polynomial images of the unit square

$$m = \frac{r}{3}, \qquad \rho^2 = \tau m^2, \tag{21}$$

$$m = \frac{1+2r}{3}, \qquad \rho^2 = \frac{r(r+2)}{3} + \tau\frac{(1-r)^2}{9}, \tag{22}$$

where $0 \leq r, \tau \leq 1$.

Let $a_n = h_1^{(n)}(0)$ in this normalization. Then

$$a_0 = 1, \qquad a_1 = a_2 = 0,$$

and

$$a_{n+3} = (1+3m)a_{n+2} - (3m^2+2m-\rho^2)a_{n+1} + (1+m)(m^2-\rho^2)a_n. \tag{23}$$

If D_N denotes $N!$ times the coefficient of degree N in the corrected curvature, then

$$D_N = \sum_{j=0}^{N} \binom{N}{j} \left(2 - \frac{N}{2} + \frac{(N-2j)^2}{2}\right) a_j a_{N-j}. \tag{24}$$

Exact conversion shows that every D_N, $0 \leq N \leq 60$, has only nonnegative tensor Bernstein coefficients on each of (21) and (22).

This finite calculation already gives a rigorous interval, not merely coefficient evidence. Normalize the Perron root to 1 and call the other two roots $0 \le u \le v \le 1$. Positive rescaling of time multiplies D_N by a positive factor and therefore preserves the Bernstein-certified signs. In the new normalization,

$$\sqrt{u} + \sqrt{v} \le 1, \qquad a_n = uv\, h_{n-3}(u, v, 1) \quad (n \ge 3),$$

where h_k is the complete homogeneous polynomial. Consequently

$$0 \le a_n \le \frac{uv}{(1-u)(1-v)} \le \frac{1}{9}.$$

For $N \ge 61$, the absolute value of (24) is therefore at most $N^2 2^N$. On $0 \le x \le 10$, where x now denotes Perron-normalized time, the tail after degree 60 is at most

$$\frac{61^2 20^{61}}{61!} \left(1 - \frac{20 \cdot 62}{61^2}\right)^{-1} < 1.$$

The Bernstein partial sum is at least its constant term 2, so the first-start inequality is proved throughout this compact time range. The calculation and the rational tail comparison are reproduced by `python/check_real_first_full.py`.

For reference, the unbounded tail has a particularly short exact form. For $0 < u < v < 1$, partial fractions give

$$h_1(t) = \frac{ve^{ut}}{(v-u)(1-u)} - \frac{ue^{vt}}{(v-u)(1-v)} + \frac{uve^{t}}{(1-u)(1-v)}.$$

Put

$$A = (1-v)t, \qquad B = (1-u)t,$$

and

$$R = \frac{1-u}{v(v-u)} e^{-A}, \qquad S = \frac{1-v}{u(v-u)} e^{-B}.$$

After factoring the last exponential, direct differentiation yields

$$-t^2 (\log h_1)'' = \frac{RA^2 + RS(B-A)^2 - SB^2}{(1-R+S)^2}. \tag{25}$$

The apparent positive mixed term is already controlled by the last term:

$$R(B-A)^2 \le B^2. \tag{26}$$

Indeed, after cancelling t^2, this follows from $v - u \le v(1-u)$ and $e^{-A} \le 1$. Thus the only part not covered by the compact certificate is proving that the right side of (25) is at most 2 for $t \ge 10$, including the confluent limits. This is now a two-exponential tail estimate on the compact root simplex, rather than a matrix inequality. The partial-fraction identity, (25), and (26) are checked symbolically by `python/check_real_first_tail_identities.py`.

There is a sharper coefficient route to the tail. Put $g_n = h_n(u, v, 1)$ and consider the normalized complete homogeneous sequence

$$q_n = \frac{g_n}{n+1}. \tag{27}$$

The remaining sequence statement is elementary, although its useful form is not the most familiar normalization.

Proposition 16 (Normalized complete-homogeneous log-convexity)**.**

$$q_n^2 \le q_{n-1}q_{n+1} \qquad (n \ge 1) \quad \textit{whenever} \quad \sqrt{u}+\sqrt{v} \le 1. \tag{28}$$

Notice that the case $n = 1$ is precisely $(1-u-v)^2 \ge 4uv$. The Jacobi–Trudi identity gives an equivalent symmetric-polynomial form:

$$n(n+2)h_n(u,v,uv) \le h_{n-1}(1,u,v)h_{n+1}(1,u,v). \tag{29}$$

Indeed, $s_{(n,n)}(1,u,v) = g_n^2 - g_{n-1}g_{n+1} = h_n(u,v,uv)$. This formulation may be more convenient for a uniform proof because it contains no quotients and its $n = 1$ case is exactly the root discriminant.

There is now stronger exact evidence for a coefficientwise proof. Introduce nonnegative x, y, r by

$$u = x^2, \qquad v = y^2, \qquad 1 = (x+y)^2 + r$$

after homogeneous rescaling, and put

$$\mathcal{D}_n(A,B,C) = (n+1)^2 h_{n-1}(A,B,C)h_{n+1}(A,B,C) - n(n+2)h_n(A,B,C)^2.$$

Exact coefficient extraction gives, through $n = 40$,

$$\mathcal{D}_n\big(x^2, y^2, (x+y)^2 + r\big) \in \mathbb{Z}_{\ge 0}[x,y,r]. \tag{30}$$

It also gives the apparently stronger monotonicity

$$\mathcal{D}_{n+1}\big(x^2,y^2,C\big) - C^2\mathcal{D}_n\big(x^2,y^2,C\big) \in \mathbb{Z}_{\ge 0}[x,y,r], \qquad C = (x+y)^2 + r. \tag{31}$$

For example,

$$\mathcal{D}_1\big(x^2, y^2, (x+y)^2 + r\big) = r(r+4xy).$$

Thus a uniform proof of either coefficient assertion would prove (28) immediately, and (31) suggests a particularly short induction. The integer coefficient calculation avoids symbolic expansion and is reproduced by `python/check_normalized_complete_cone.py`.

We now prove the proposition. By continuity, order the roots as $0 < u < v < 1$; the confluent and boundary cases follow by taking limits. Let X be uniform on $[v,1]$ and Y uniform on $[u,v]$, and write $A_j = \mathbb{E}X^j$ and $B_j = \mathbb{E}Y^j$. Partial fractions followed by one integration gives

$$q_n = \frac{A_n - uB_n}{1-u}. \tag{32}$$

Indeed, the representing signed density vanishes on $[0,u]$, equals $-u(1-v)/(v-u)$ on $[u,v]$, and equals 1 on $[v,1]$, before division by $(1-u)(1-v)$. Consequently

$$\begin{aligned} (1-u)^2(q_{n-1}q_{n+1} - q_n^2) &= \Delta_A + u^2\Delta_B - u\Gamma, \\ \Delta_A = A_{n-1}A_{n+1} - A_n^2, &\qquad \Delta_B = B_{n-1}B_{n+1} - B_n^2, \\ \Gamma &= \mathbb{E}\big[X^{n-1}Y^{n-1}(X-Y)^2\big]. \end{aligned} \tag{33}$$

Here both Δ terms are nonnegative. The case $n = 1$ is the root discriminant noted above. For $n \ge 2$, pointwise maximization in Y gives

$$\Gamma \le \min\left\{ v^{n-1}, \frac{4(n-1)^{n-1}}{(n+1)^{n+1}} \right\} A_{n+1}. \tag{34}$$

Since the root condition implies $u \le \min\{v, (1-\sqrt{v})^2\}$, it remains only to prove the one-variable inequality

$$\min\{v, (1-\sqrt{v})^2\} \min\left\{ v^{n-1}, \frac{4(n-1)^{n-1}}{(n+1)^{n+1}} \right\} A_{n+1} \le \Delta_A. \tag{35}$$

Put

$$c_n = \frac{4(n-1)^{n-1}}{(n+1)^{n+1}}, \qquad A_j = \frac{1-v^{j+1}}{(j+1)(1-v)}.$$

Direct simplification gives

$$\Delta_A = \frac{(1-v)^2 P_n(v)}{n(n+1)^2(n+2)}, \tag{36}$$

where

$$P_n(v) = \sum_{k=0}^{n-2} \binom{k+3}{3} \left(v^k + v^{2n-2-k}\right) + \binom{n+2}{3} v^{n-1}. \tag{37}$$

Thus $P_n(v) \ge 1$. Pairing the terms in each parenthesis by AM–GM and using $\sum_{k=0}^{n-2} \binom{k+3}{3} = \binom{n+2}{4}$ also gives

$$P_n(v) \ge \frac{n(n+1)^2(n+2)}{12}\, v^{n-1}. \tag{38}$$

First suppose $0 \le v \le 1/4$ and $n \ge 6$. Since $A_{n+1} \le ((n+2)(1-v))^{-1}$, the left side of (35) is at most $v^n/((n+2)(1-v))$. Equations (36) and $P_n \ge 1$ reduce the claim to

$$n(n+1)^2 v^n \le (1-v)^3.$$

It is enough to set $v = 1/4$. The resulting inequality holds already at $n = 4$, and its left side decreases with n.

Now suppose $1/4 \le v \le 1$, write $b = \sqrt{v}$, and take $n \ge 6$. By (38), it is enough to show

$$\frac{12 A_{n+1}}{(1+b)^2} \min\left\{1, \frac{c_n}{b^{2n-2}}\right\} \le 1. \tag{39}$$

Set

$$b_n = c_n^{1/(2n-2)} = \sqrt{\frac{n-1}{n+1}} \left(\frac{2}{n+1}\right)^{1/(n-1)}.$$

For $1/2 \le b \le b_n$, use $A_{n+1} \le ((n+2)(1-b^2))^{-1}$. Since $(1-b)(1+b)^3$ decreases for $b \ge 1/2$, it suffices that

$$(n+2)(1-b_n)(1+b_n)^3 \ge 12. \tag{40}$$

For $b \ge b_n$, the left side of (39) is nonincreasing. Indeed,

$$A_{n+1} = \frac{1}{n+2} \sum_{j=0}^{n+1} b^{2j},$$

and $bA'_{n+1}/A_{n+1} \le n+1$, because the weights b^{2j} decrease with j. This is at most $2n-2+2b/(1+b)$, the logarithmic derivative with respect to $\log b$ of $b^{2n-2}(1+b)^2$. Thus this region also reduces to (40).

For completeness, (40) has a short calculus check. Extend b_n to $b(x)$ for real $x \ge 6$. Then

$$\frac{d}{dx} \log b(x) = \frac{\log((x+1)/2)}{(x-1)^2} > 0.$$

Moreover, $(x+2)(1-b(x))$ is increasing. To see this, use $b^{-1}-1 \geq -\log b$ and

$$\frac{(x-1)^2}{2}\log\frac{x+1}{x-1} \geq x-2+\frac{1}{x} \geq 3\log\frac{x+1}{2} \qquad (x \geq 6).$$

Hence $(x+2)(1-b(x))(1+b(x))^3$ is increasing. Finally $b(6) < 2/3$, so its value at $x=6$ is greater than $8(1-2/3)(1+2/3)^3 = 1000/81 > 12$.

It remains only to cover $n = 2,3,4,5$. On the two intervals $0 \leq b \leq 1/2$ and $1/2 \leq b \leq 1$, exact Bernstein conversion proves the stronger inequality

$$\Delta_A \geq \min\{b^2, (1-b)^2\}\, c_n A_{n+1}.$$

The rational certificates are reproduced by `python/check_uniform_moment_bound.py`. This proves (35); inserting it into (33) proves the proposition.

Here is why the proposition finishes the tail. Write $w = h_1 - 1$ and

$$t^2\{ww'' - (w')^2\} + 2w^2 = (uv)^2 \sum_{N\geq 6} \widehat{D}_N \frac{t^N}{N!}.$$

Since $w = uv\sum_{N\geq 3} g_{N-3}t^N/N!$, one has, with $\kappa_N(j) = 2 - N/2 + (N-2j)^2/2$,

$$\widehat{D}_N = \sum_{j=3}^{N-3} \binom{N}{j} \kappa_N(j) g_{j-3} g_{N-j-3}.$$

Set $M = N-6$ and

$$B_i = \binom{N}{i+3}(i+1)(M-i+1), \qquad 0 \leq i \leq M.$$

The summand weight is $B_i q_i q_{M-i}$. By (28), the symmetric multiplier $q_i q_{M-i}$ is nondecreasing with $|i - M/2|$. The same is true of $\kappa_N(i+3)$. The weighted Chebyshev inequality therefore compares its average with the one obtained by setting $q_i = 1$. The numerator of that baseline average is

$$\sum_{i=0}^{M} B_i \kappa_N(i+3) = \frac{2^N(N^2-17N+32) + 4N^4 - 24N^3 + 68N^2 - 48N - 64}{4}, \tag{41}$$

which is positive for every $N \geq 15$. Hence the proposition implies $\widehat{D}_N \geq 0$ for all $N \geq 15$.

The finitely many lower coefficients already combine positively in the tail range. With $y = 10/t$, exact tensor Bernstein conversion on $0 \leq s, d, y \leq 1$, where

$$\sqrt{u} = \frac{s(1-d)}{2}, \qquad \sqrt{v} = \frac{s(1+d)}{2},$$

proves

$$t^{-16}\sum_{N=6}^{16} \widehat{D}_N \frac{t^N}{N!} \geq 0 \qquad (t \geq 10). \tag{42}$$

Thus (28) implies $V \geq 1$ on the whole remaining tail, and (18) gives $E \leq \theta(2-(1-\theta)L^2) \leq 2$. The exact low block, the signs of $\widehat{D}_N$ through $N = 40$, and the first twenty cases of the sequence lemma are reproduced by `python/check_real_first_variance_tail.py`.

For comparison, a second possible reduction is suggested by scale monotonicity. Put

$$\sqrt{u} = \frac{s(1-d)}{2}, \qquad \sqrt{v} = \frac{s(1+d)}{2}, \qquad 0 \le s, d \le 1.$$

Numerical differentiation indicates that E is nondecreasing in s whenever $E \ge 0$. Such a conditional monotonicity would reduce the tail to $s = 1$, which is exactly the repeated-rate boundary $a = b$. The observed maximum on that boundary is approximately 1.30148. These observations are reproduced by `python/explore_real_first.py`; they are evidence, not part of the proof.

The complex-pair region itself admits a compact two-parameter normalization. This removes the three original rates from a direct analytic attack on the first start. Write the roots of P as $\lambda, \mu \pm i\nu$, put $d = \lambda - \mu > 0$, and set

$$x = dt, \qquad m = \frac{\mu}{d}, \qquad \omega = \frac{\nu}{d}.$$

Comparison of the coefficients of P gives

$$a + b = d(1 + 3m), \qquad ab = d^2(3m^2 + 2m + \omega^2), \qquad Q = d^3(1+m)(m^2+\omega^2).$$

Consequently the admissible parameter set is

$$-\frac{1}{3} \le m \le \frac{1}{3}, \qquad \max\{0, -3m^2 - 2m\} \le \omega^2 \le \frac{1 - 2m - 3m^2}{4}, \tag{43}$$

with degenerate boundary points understood by continuity. After the harmless exponential shift $e^{-\mu t}$, the first component has the explicit form

$$e^{-\mu t} h_1(t) = \frac{H_{m,\omega}(x)}{1+\omega^2}, \tag{44}$$

where

$$H_{m,\omega}(x) = (m^2+\omega^2)e^x + (1-m^2)\cos(\omega x) - \frac{(m+1)(m+\omega^2)}{\omega}\sin(\omega x).$$

Thus the first bound in the complex-pair region is exactly the elementary three-variable inequality

$$x^2\big(H_{m,\omega}H''_{m,\omega} - (H'_{m,\omega})^2\big) + 2H^2_{m,\omega} \ge 0 \tag{45}$$

for $x \ge 0$ and the compact parameter domain (43). This is an equivalent reduction, not yet a proof, but it supplies a second concrete target with no matrix or Laplace-transform variables left.

The Wronskian in (45) admits a further exact factorization. This removes the second derivatives and isolates the only term which can be negative. Put

$$z = \omega^2, \qquad A = m^2 + z, \qquad T = (1-m^2)\cos(\omega x) - (m+1)(m+z)S_\omega(x),$$

so that $H_{m,\omega} = Ae^x + T$. The trigonometric part is a harmonic oscillator, and direct simplification gives

$$T'^2 + zT^2 = A(m+1)^2(1+z), \tag{46}$$

$$(1-z)T - 2T' = (m+1)(1+z)\left((m+1)\cos(\omega x) + (z-m)S_\omega(x)\right). \tag{47}$$

Consequently

$$H_{m,\omega}H''_{m,\omega} - (H'_{m,\omega})^2 = A(m+1)(1+z)\big[e^x\{(m+1)\cos(\omega x) + (z-m)S_\omega(x)\} - (m+1)\big]. \tag{48}$$

Thus the complex first-start target is the derivative-free inequality

$$2(Ae^x+T)^2+x^2A(m+1)(1+z)\big[e^x\{(m+1)\cos(\omega x)+(z-m)S_\omega(x)\}-(m+1)\big]\geq 0. \tag{49}$$

All factors outside the square brackets are nonnegative on (43). At the confluent boundary $z=0$ this has the continuous specialization

$$H_{m,0}(x)=m^2e^x+1-m^2-m(m+1)x, \qquad 0\leq m\leq \frac{1}{3},$$

and the bracket in (49) becomes $e^x(m+1-mx)-(m+1)$. These identities are checked symbolically by `python/check_complex_first_factorization.py`.

The confluent boundary just displayed can in fact be settled completely. This is one of the two lower boundary arcs suggested by (49).

Proposition 17 (Confluent complex boundary)**.** *For $z=0$, $0\leq m\leq 1/3$, and $x\geq 0$, the first-start inequality* (45) *holds.*

Proof. The case $m=0$ is immediate, so suppose $m>0$ and write $r=1+m$. With

$$H=m^2e^x+1-m^2-mrx, \qquad J=e^x(r-mx)-r,$$

the left side of (45) is

$$D=2H^2+x^2m^2rJ. \tag{50}$$

If $J\geq 0$ there is nothing to prove. Since $J'=e^x(1-mx)$ and $J(0)=0$, the alternative $J<0$ forces $x>1/m$.

We use twice the elementary bound

$$\frac{m^2e^{1/m}}{1+m}\geq\frac{e^3}{12}>\sqrt{2} \qquad (0<m\leq 1/3).$$

Indeed, the expression on the left is decreasing on this interval. It first implies, for $x\geq 1/m$,

$$H\geq\frac{1}{2}m^2e^x.$$

If moreover $x\leq r/m$, then $-J\leq r$ and $me^x\geq\sqrt{2}\,xr$; hence

$$D\geq\frac{1}{2}m^4e^{2x}-x^2m^2r^2\geq 0.$$

It remains to consider $x\geq r/m$. Regard the right side of (50) as the quadratic

$$\mathcal{D}(m,x,Y)=2(m^2Y+1-m^2-mrx)^2+x^2m^2r\{Y(r-mx)-r\}.$$

Put $u=mx-r\geq 0$. Then

$$m=\frac{u+1}{x-1}, \qquad x=3u+4+v$$

for some $v\geq 0$, because $m\leq 1/3$. Also $x\geq 4$ and $e^x\geq x^4/6$: the quotient e^x/x^4 is increasing there, and the inequality at $x=4$ follows from the first six terms of the exponential series. Direct expansion after the preceding substitution shows that both

$$72(x-1)^4\mathcal{D}\left(m,x,\frac{x^4}{6}\right), \qquad 6(x-1)^4\partial_Y\mathcal{D}\left(m,x,\frac{x^4}{6}\right)$$

are polynomials in u,v with nonnegative integer coefficients. Therefore $\mathcal{D}$ is nonnegative at $Y=x^4/6$ and is increasing for all larger Y. Taking $Y=e^x$ proves the result. The two polynomial expansions (respectively 80 and 35 monomials) are verified exactly by `python/check_complex_first_factorization.py`. □

The other lower boundary arc can also be proved. The following argument uses an exact finite Bernstein certificate only on a compact interval; the tail is handled analytically.

Proposition 18 ($a = 0$ complex boundary)**.** *For $m = -q$, $0 \le q \le 1/3$, and $z = 2q - 3q^2$, the first-start inequality* (45) *holds for every $x \ge 0$.*

Proof. Undo the harmless exponential shift and put

$$h(x) = e^{-qx} \frac{H_{-q,\sqrt{z}}(x)}{1+z}.$$

In the ordered canonical coordinates this is the first component with

$$b = 1 - 3q, \qquad Q = 2q(1-q)^2,$$

and hence

$$h''' = (1-3q)h'' + 2q(1-q)^2 h, \qquad h(0) = 1, \quad h'(0) = h''(0) = 0. \tag{51}$$

Multiplication by e^{-qx} does not change logarithmic curvature, so it is enough to prove

$$D_q(x) = x^2\{hh'' - (h')^2\} + 2h^2 \ge 0.$$

We first treat $0 \le x \le 10$. Write $a_n(q) = h^{(n)}(0)$. Equation (51) gives the exact polynomial recurrence

$$a_{n+3} = (1-3q)a_{n+2} + 2q(1-q)^2 a_n, \qquad a_0 = 1, \quad a_1 = a_2 = 0.$$

If $D_N(q)$ is $N!$ times the coefficient of x^N in $D_q(x)$, then

$$D_N(q) = \sum_{j=0}^{N} \binom{N}{j} \left(2 - \frac{N}{2} + \frac{(N-2j)^2}{2}\right) a_j(q) a_{N-j}(q).$$

Exact rational calculation shows that, for every $0 \le N \le 80$, $D_N(x/3)$ has nonnegative coefficients in the Bernstein basis on $0 \le x \le 1$. Thus the degree-80 Taylor polynomial of D_q is at least its constant term 2 throughout the required parameter interval.

For completeness, the Taylor tail has a short uniform bound. The spectral form is

$$h = c_0 e^{(1-q)x} + c_+ e^{(-q+i\sqrt{z})x} + c_- e^{(-q-i\sqrt{z})x}, \qquad c_- = \overline{c_+}, \tag{52}$$

where

$$c_0 = \frac{2q(1-q)}{1+z}.$$

All three exponents have modulus at most 1, and $|c_0| + |c_+| + |c_-| \le 2$. For a sum $h = \sum c_r e^{\lambda_r x}$ one has

$$D_q(x) = \sum_{r,s} c_r c_s \left\{2 + \frac{x^2}{2}(\lambda_r - \lambda_s)^2\right\} e^{(\lambda_r + \lambda_s)x}.$$

Consequently, on $0 \le x \le 10$, the absolute remainder after total degree 80 is at most

$$8e^{20}\left\{\frac{20^{81}}{81!} + 100\frac{20^{79}}{79!}\right\} < 1.$$

Here one may use the elementary bound $e^{20} < 500000000$; the final inequality is rational arithmetic. The exact Bernstein calculation and this tail check are reproduced by `python/check_complex_first_a0_boundary.py`. It follows that $D_q(x) > 1$ on this compact interval.

It remains to give a purely analytic estimate for $x \geq 10$. Put

$$U = \cos(\sqrt{z}, x) + 3qS_{\sqrt{z}}(x).$$

On the present boundary the square bracket in (49) is $(1-q)(e^x U - 1)$. If it is nonnegative, the result is immediate. Otherwise $U < e^{-x} < 1/2$. If $\sqrt{z}, x \leq 1$, then $\cos(\sqrt{z}, x) \geq 1 - zx^2/2 \geq 1/2$ and the sine term in U is nonnegative, a contradiction. Hence $zx^2 > 1$. Since $z \leq 2q$, this implies

$$q > \frac{1}{2x^2}, \qquad c_0 \geq q > \frac{1}{2x^2}. \tag{53}$$

Let $R = |c_+| + |c_-|$. From the cosine and sine coefficients in (52),

$$R \leq \frac{1-q^2}{1+z} + \frac{q(1-q)(1-3q)}{\sqrt{z}(1+z)} \leq \frac{5}{4}.$$

For the last inequality, the first summand is at most 1, while the square of the assertion that the second is at most $1/4$ reduces to

$$171q^3 - 96q^2 + 7q - 2 \leq 0.$$

This follows from $171q^3 \leq 57q^2$ and the negative-discriminant quadratic $-39q^2 + 7q - 2$.

The leading–oscillatory spectral gaps have squared modulus at most $4/3$. Therefore, with $L = 2 + 2x^2/3$, formula (52) gives

$$D_q(x) \geq e^{-2qx}\big[2c_0^2 e^{2x} - L\{2c_0 R e^x + R^2\}\big].$$

For $x \geq 10$ the bracket is increasing as a function of $c_0 \geq 1/(2x^2)$: its derivative is nonnegative because $e^x/x^2 \geq LR$, which follows already from $L \leq 7x^2/10$, $R \leq 5/4$, and $e^{10} > 22000$. Using the same bounds, its value at the lower endpoint is at least

$$\frac{e^{2x}}{2x^4} - \frac{7}{8}e^x - \frac{35}{32}x^2 > 0.$$

Indeed, after division by e^x, the first term is increasing and the last is decreasing for $x \geq 10$; at $x = 10$ the claim follows already from $e^{10} > 22000$. This proves the tail and completes the boundary. □

In fact the same finite-certificate strategy, with a uniform complex-analytic remainder estimate, fills the interior of the complex-root chart.

Proposition 19 (Complex-pair first start)**.** *For every parameter pair in* (43) *and every $x \geq 0$, the first-start inequality* (45) *holds.*

Proof. We first prove the assertion on $0 \leq x \leq 10$. Two overlapping choices of the harmless exponential shift keep the exact certificates small. For $m \geq 0$, use

$$F = \frac{H_{m,\omega}}{1+z}.$$

Its derivatives at zero satisfy

$$f_0 = 1, \qquad f_1 = -m, \qquad f_2 = m^2, \qquad f_{n+3} = f_{n+2} - zf_{n+1} + zf_n.$$

Let P_{70} be the degree-70 Taylor polynomial of $x^2\{FF'' - (F')^2\} + 2F^2$. Parameterize the positive half of (43) by

$$m = \frac{r}{3}, \qquad z = \tau\frac{3 - 2r - r^2}{12}, \qquad 0 \le r, \tau \le 1,$$

and put $x = 10u$, $0 \le u \le 1$. Exact conversion of $P_{70}(10u)$ to the tensor Bernstein basis in (r, τ, u) gives only positive coefficients; the smallest is

$$\frac{7445875529}{6372236871} > 1. \tag{54}$$

For $m \le 0$, use instead $h = e^{mx}F$, the unshifted canonical component. Its derivatives satisfy

$$h_{n+3} = (1 + 3m)h_{n+2} - (3m^2 + 2m + z)h_{n+1} + (1 + m)(m^2 + z)h_n, \qquad h_0 = 1, \qquad h_1 = h_2 = 0.$$

Write $r = -3m$ and parameterize this half by

$$z = \frac{r(2 - r)}{3} + \tau\frac{(1 - r)^2}{4}, \qquad 0 \le r, \tau \le 1.$$

For every Taylor degree $0 \le N \le 70$, the corresponding curvature coefficient has only nonnegative tensor Bernstein coefficients in (r, τ). Hence the degree-70 partial sum is at least its constant term 2 for $x \ge 0$.

The omitted tail is uniformly smaller than 1 in both cases. Indeed, the chosen function (F for $m \ge 0$ and h for $m \le 0$), together with its first two derivatives, has modulus at most $36e^{34}$ on the circle $|x| = 34$. This follows directly from the exponential–sine–cosine formula: all relevant spectral frequencies have modulus at most 1, all scalar coefficients have modulus at most 1, and $|S_\omega(x)| \le |x|e^{|\omega x|}$. Thus the curvature has modulus at most

$$M = (2 \cdot 34^2 + 2)(36e^{34})^2.$$

Cauchy's estimate bounds its Taylor tail on $|x| \le 10$ by

$$M\frac{(5/17)^{71}}{1 - 5/17} < 1;$$

the last inequality follows already from $e^{34} < 6 \cdot 10^{14}$ and is an exact rational check. The two Bernstein certificates and the tail bound are verified by `python/check_complex_first_full.py`. Combined with (54), they prove the claim for $0 \le x \le 10$.

It remains to give a uniform analytic argument for $x \ge 10$. Put

$$A = m^2 + z, \qquad U = \cos(\omega x) + \frac{z - m}{m + 1}S_\omega(x), \qquad y = x\sqrt{A}.$$

The square bracket in (49) is $(m + 1)(e^x U - 1)$. If it is nonnegative, there is nothing to prove. Suppose therefore that $U < e^{-x}$. The parameter domain gives

$$z \le \frac{1}{3}, \qquad A \le \frac{4}{9}, \qquad (z - m)^2 \le A, \qquad m + 1 \ge \frac{2}{3}.$$

Using $\cos v \ge 1 - v^2/2$ and $|S_\omega(x)| \le x$, we obtain

$$U \ge 1 - \frac{y^2}{2} - \frac{3y}{2}.$$

Since $e^{-x} < 1/8$, this forces $y > 1/2$.

The trigonometric term in $H = Ae^x + T$ has the useful identity

$$T = (1 - m^2)U - 2AS_\omega(x). \tag{55}$$

If $0 \le U < e^{-x}$, then $H \ge A(e^x - 2x) \ge (9/10)Ae^x$. If $U < 0$, put

$$G(y) = \frac{y^2}{2} + \frac{3y}{2} - 1.$$

The preceding lower bound gives $|U| \le G(y)$, while

$$G(y) \le \frac{17}{16}y^2,$$

because the difference after division by y^2 is $(3y-4)^2/(16y^2)$. Equation (55) then gives

$$H \ge Ae^x - G(y) - 2Ax \ge \frac{9}{10}Ae^x;$$

the last step uses $x \ge 10$, $y > 1/2$, and $e^{10} > 22000$.

Finally, $(m+1)^2(1+z) \le 64/27$. In the case $U \ge 0$, the possibly negative bracket has magnitude at most 1, and hence

$$D_H \ge \frac{3}{2}A^2e^{2x} - \frac{64}{27}x^2A > 0.$$

In the case $U < 0$, use $1 + e^x|U| \le 1 + (17/16)e^x y^2$ to obtain

$$D_H \ge y^4\left\{\frac{3e^{2x}}{2x^4} - \frac{68}{27}e^x - \frac{256}{27}\right\} > 0.$$

Both final inequalities follow at $x = 10$ from $y > 1/2$ and $e^{10} > 22000$, and their left sides only improve thereafter. This proves the full complex-pair region. □

There is also a natural, but no longer necessary, monotonicity route from the two boundaries into the interior of the complex-root domain. Normalize $\widehat{H} = H_{m,\omega}/(1+z)$. Its Laplace transform with respect to x and its z-derivative are

$$\begin{aligned}\mathcal{L}\widehat{H}(s) &= \frac{s^2 - (m+1)s + m^2 + m + z}{(s-1)(s^2+z)},\\ \partial_z\mathcal{L}\widehat{H}(s) &= \frac{(m+1)(s-m)}{(s-1)(s^2+z)^2}.\end{aligned} \tag{56}$$

Equivalently,

$$\partial_z\widehat{H} = (m+1)\,e^x * \big(\cos(\omega x) - mS_\omega(x)\big) * S_\omega(x).$$

Exact numerical minimization indicates that the normalized curvature in (45), divided by $(1+z)^2$, is nondecreasing in z for fixed (x, m). This monotonicity is no longer needed for the proof, and its individual Taylor coefficients do change sign. Formula (56) is a compact starting point for that stronger comparison; at present it remains conjectural.

The missing middle bound has an equally explicit form in the same chart. Let

$$\delta = \sqrt{1 - 2m - 3m^2 - 4\omega^2}, \qquad \beta = \frac{1+m+\delta}{2} = \frac{b-\mu}{d}.$$

Then, up to the positive constant x_2/d,

$$e^{-\mu t}h_2(t) = \frac{K_{m,\omega}(x)}{1+\omega^2},$$

where

$$K_{m,\omega}(x) = (1-\beta)\big(e^x - \cos(\omega x)\big) + \frac{\omega^2+\beta}{\omega}\sin(\omega x).$$

Thus the sharper middle target in (15) is equivalent in the complex-pair region to

$$x^2\big(K_{m,\omega}K''_{m,\omega} - (K'_{m,\omega})^2\big) + K^2_{m,\omega} \geq 0. \tag{57}$$

The quotients by ω in both formulas have their continuous meanings at $\omega = 0$.

The parameter dependence of the last inequality can be reduced further. Set

$$A = 1-\beta = \frac{\lambda - b}{d}, \qquad \rho = \sqrt{1-3\omega^2}.$$

Eliminating m and b from (43) gives

$$0 \leq \omega^2 \leq \frac{1}{3}, \qquad \frac{2(1-\rho)}{3} \leq A \leq \frac{2-\rho}{3}. \tag{58}$$

The left endpoint is precisely $a = 0$, and the right endpoint is precisely $a = b$. Moreover, with

$$S_\omega(x) = \frac{\sin(\omega x)}{\omega}, \qquad D_\omega(x) = e^x - \cos(\omega x) - S_\omega(x),$$

one has the affine formula

$$K_{m,\omega}(x) = (1+\omega^2)S_\omega(x) + AD_\omega(x). \tag{59}$$

For fixed (x,ω), the left side of (57) is a quadratic polynomial in A. The following argument proves that it is nonnegative on the whole interval, not merely at its endpoints.

Proposition 20 (Complex-pair middle start)**.** *For every parameter pair in* (58) *and every $x \geq 0$, the stronger middle inequality* (57) *holds.*

Proof. Put $z = \omega^2$ and $B = 1 - A + z$. Then

$$K = Ae^x - A\cos(\omega x) + B\frac{\sin(\omega x)}{\omega}.$$

Let C_N be $N!$ times the coefficient of x^N in

$$\mathcal{C}(x) = K^2 + x^2(KK'' - (K')^2).$$

For $N \geq 3$, set $M = N(N-1)/2$ and

$$R_N = (1+i\omega)^N + M(1-i\omega)^2(1+i\omega)^{N-2}, \qquad J_N = \frac{\operatorname{Im} R_N}{\omega}.$$

Writing the trigonometric part as a pair of conjugate exponentials gives the exact identity

$$C_N = A^2 2^N + 2\operatorname{Re}\left[\left(\frac{-A - iB/\omega}{2}\right)^2 (2i\omega)^N\right] - 2A^2 \operatorname{Re} R_N + 2ABJ_N. \tag{60}$$

All expressions at $\omega = 0$ are interpreted by continuity.

The domain has the useful elementary consequences

$$0 \le z \le A, \qquad 0 \le B \le 1, \qquad r := \sqrt{1+z} \le \frac{2}{\sqrt{3}}. \tag{61}$$

Moreover,

$$J_N(0) = N + M(N-4) > 0 \quad (N \ge 4).$$

The polynomial $R_N(\omega)$ is a sum of a product of N linear factors and M times another such product. Taylor's theorem, using the conjugation symmetry at the origin, therefore gives

$$J_N \ge J_N(0) - \frac{z}{6}(1+M)N(N-1)(N-2)r^{N-3}. \tag{62}$$

Indeed, the absolute value of the third derivative of either product is at most $N(N-1)(N-2)r^{N-3}$ on $[0,\omega]$.

We now use (62) in (60), retain the nonnegative term $2ABJ_N(0)$, and bound the other two real parts by their moduli. Since $z \le A$ and $B \le 1$, for $N \ge 6$ this yields

$$\begin{aligned} C_N \ge A^2\Bigg\{ & 2^N - 2(1+M)r^N - \frac{1}{3}(1+M)N(N-1)(N-2)r^{N-3} \\ & - 2^{N-1}\left(3^{-N/2} + 3^{3-N/2}\right)\Bigg\}. \end{aligned} \tag{63}$$

For example, the conjugate-pair term is bounded by

$$2^{N-1}\left(A^2 z^{N/2} + B^2 z^{N/2-1}\right) \le 2^{N-1}A^2\left(3^{-N/2} + 3^{3-N/2}\right).$$

The braces in (63) are positive for every $N \ge 25$. After division by 2^N, their total subtracted part is

$$E_N = \left(2(1+M) + 14 + \frac{\sqrt{3}}{8}(1+M)N(N-1)(N-2)\right)3^{-N/2}.$$

At $N = 25$ its numerator is $616 + 519225\sqrt{3}$, whereas its denominator is $531441\sqrt{3}$, so $E_{25} < 1$. Thereafter each of the two positive numerator summands grows by a factor less than $4/3$, while the denominator grows by $\sqrt{3}$. Indeed, their successive ratios are bounded by their values at $N = 25$, namely $666/616$ and $(326/301)(26/23)$, both less than $4/3$. Thus E_N decreases for $N \ge 25$.

It remains only to check the finite range. Parametrize the complete domain by

$$\omega^2 = \frac{1-\rho^2}{3}, \qquad A = \frac{2-2\rho+\rho y}{3}, \qquad 0 \le \rho, y \le 1.$$

For $N = 4, \dots, 24$, direct power-to-Bernstein conversion of C_N gives only nonnegative rational coefficients, first in y and then in ρ. This is a finite exact certificate: the derivative sequence used to form it is

$$K^{(n)}(0) = A(1 - c_n z^{\lfloor n/2 \rfloor}) + B s_n z^{\lfloor (n-1)/2 \rfloor},$$

where $(c_n) = (1, 0, -1, 0)$ and $(s_n) = (0, 1, 0, -1)$ periodically. The short exact-arithmetic verification is `python/check_complex_middle_boundaries.py`. Finally, $C_0 = C_1 = C_2 = C_3 = 0$. Hence every Taylor coefficient of $\mathcal{C}$ is nonnegative, proving the result for $x \ge 0$. □

Thus the middle start is now complete in both the real- and complex-spectrum regions. By the convolution argument preceding (16), the third start is complete as well. At this stage, only the first canonical start remains open.

The calculation in (18) in fact has a root-free extension which proves a substantial part of the remaining case. Assume $Q > 0$ and define

$$\mathcal{M}(t) = \frac{h_1(t) - 1}{Qt^3}.$$

Since $h_1' = ch_3$, the function $\mathcal{M}$ is, up to a positive constant, $t^{-3}\int_0^t h_3(s)\,ds$. The completed third-start result and the convolution lemma imply that $\mathcal{M}$ is log-convex. Its series at the origin gives

$$(\log \mathcal{M})'(0) = \frac{a+b}{4} \geq 0;$$

hence $(\log \mathcal{M})' \geq 0$ everywhere. Put

$$\vartheta = \frac{h_1 - 1}{h_1}, \qquad \mathcal{L} = 3 + t(\log \mathcal{M})', \qquad \mathcal{V} = t^2(\log \mathcal{M})''.$$

Then $\mathcal{L} \geq 3$, $\mathcal{V} \geq 0$, and direct differentiation gives

$$-t^2(\log h_1)'' = \vartheta\big(3 - \mathcal{V} - (1-\vartheta)\mathcal{L}^2\big). \tag{64}$$

It follows immediately that the first-start target holds, for every spectrum, throughout the region

$$h_1(t) - 1 \leq 5 + 3\sqrt{3}. \tag{65}$$

Indeed, this is equivalent to $\vartheta \leq (1+\sqrt{3})/3$, and the right side of (64) is then at most $\vartheta(9\vartheta - 6) \leq 2$. Outside (65), the sole remaining scalar inequality is

$$\mathcal{V} + (1-\vartheta)\mathcal{L}^2 \geq 3 - \frac{2}{\vartheta}. \tag{66}$$

For real roots, $\mathcal{V}$ is precisely the tilted-simplex variance V in (19); for a complex pair it remains a nonnegative convolution curvature. Thus (66) unifies the two spectral regions.

Exact computation supports the cleaner, stronger inequality

$$\mathcal{V} + (1-\vartheta)\mathcal{L}^2 \geq \vartheta. \tag{67}$$

It implies (66), since $\vartheta \geq 3 - 2/\vartheta$ for $0 < \vartheta \leq 1$. In the original variables it is equivalent to

$$t^2\big(h_1 h_1'' - (h_1')^2\big) + 2h_1^2 - h_1 - 1 \geq 0, \tag{68}$$

and would give the sharper bound $E_1 \leq \vartheta(3-\vartheta) \leq 2$. At coefficient level, (68) replaces $\Gamma(C,J,K)$ by $\Gamma(C,J,K) - \alpha(C,J,K)$. Exact tests find the same five negative triples $(2,0,K)$, $3 \leq K \leq 7$, and no others. Grouping with the neighboring $C = 1, 3$ terms gives positive quadratics whose discriminants are

$$-5907200, \quad -1402143, \quad -7154704640, \quad -11951446784, \quad -263131073024.$$

Thus the two normalized coefficient monotonicities below would prove even the strengthened form (68); no additional infinite family is introduced.

The completed third bound almost closes the first one by a barrier argument. This gives a useful quantitative target for strengthening the convolution step. Introduce logarithmic time $\tau = \log t$ and the scaled variables

$$Z = tz, \qquad P = tp, \qquad R = tq, \qquad A_0 = ta, \qquad B_0 = tb.$$

Then (14) becomes

$$\dot{Z} = Z(1 + B_0 + R - Z), \qquad \dot{P} = P(1 + Z - A_0 - P), \qquad \dot{R} = R(1 + A_0 + P - B_0 - R),$$

where a dot denotes $d/d\tau$. Set

$$E_1 = Z(Z - B_0 - R), \qquad E_3 = R(B_0 + R - A_0 - P).$$

A direct differentiation, followed only by the substitution $E_1 = 2$, gives the exact first-contact identity

$$\dot{E}_1\Big|_{E_1=2} = Z(E_3 - 2) + 4 - \frac{4}{Z}. \tag{69}$$

At such a contact one has $Z > \sqrt{2}$. The established third inequality $E_3 \le 2$ alone therefore falls just short of forcing $\dot{E}_1 \le 0$. It would be enough to prove the following quantitative slack implication:

$$E_1 = 2 \quad \Longrightarrow \quad 2 - E_3 \ge \frac{4(Z-1)}{Z^2}. \tag{70}$$

The left side is exactly $t^2(\log(t^{-2}h_3))''$, so (70) asks for a quantitative version of the same convolution log-convexity argument that already proves its nonnegativity.

That slack itself has an exact positive decomposition. Put

$$F(s) = \frac{e^{-bs}h_2(s)}{s}, \qquad G(t) = \frac{e^{-bt}h_3(t)}{t^2} = x_3 \int_0^1 uF(tu)\,du,$$

and let μ_t be the probability measure on $[0,1]$ with density proportional to $uF(tu)$. If

$$E_2(s) = s^2 p(s)(a + p(s) - z(s)), \qquad L(s) = s(\log F)'(s),$$

then differentiation under the integral gives

$$2 - E_3(t) = \mathbb{E}_{\mu_t}[1 - E_2(tu)] + \mathrm{Var}_{\mu_t}(L(tu)). \tag{71}$$

Both terms are nonnegative by the stronger middle bound. Combining (70) and (71) isolates the remaining dynamic question: at a first contact, one must extract a lower bound on either the averaged middle slack or the spread of the logarithmic slopes. No oscillatory root variables occur in this formulation.

There is a complementary coefficient reduction for the first canonical start. It sharply localizes the cancellation that a proof still has to explain. Write

$$S = a + b, \qquad \Delta = (a - b)^2, \qquad Q = cx_2x_3.$$

The ordinary generating function of the return-walk weights is

$$\begin{aligned}\sum_{n\ge 0}(B^n)_{11}z^n &= \frac{(1-az)(1-bz)}{(1-az)(1-bz) - Qz^3} \\ &= \sum_{C\ge 0}\frac{Q^C z^{3C}}{\big((1 - Sz/2)^2 - \Delta z^2/4\big)^C}.\end{aligned} \tag{72}$$

Here the term with $C = 0$ is 1. For $C \geq 1$, expansion of the last display gives the manifestly nonnegative coefficient

$$\alpha(C, J, K) = \binom{C + J - 1}{J} \binom{2C + 2J + K - 1}{K} 2^{-2J-K} \tag{73}$$

of $Q^C \Delta^J S^K z^{3C+2J+K}$.

This turns coefficientwise curvature into a finite binomial-sum question. Set $\alpha(0, 0, 0) = 1$, with all other $\alpha(0, J, K) = 0$, put $N = 3C + 2J + K$, and define

$$\begin{aligned} \Gamma(C, J, K) = \sum_{\substack{0 \leq c \leq C \\ 0 \leq j \leq J \\ 0 \leq k \leq K}} \binom{N}{3c + 2j + k} \left(2 - \frac{N}{2} + \frac{(N - 2(3c + 2j + k))^2}{2}\right) \\ \times \alpha(c, j, k)\alpha(C - c, J - j, K - k). \end{aligned} \tag{74}$$

Thus $\Gamma(C, J, K)$ is exactly the coefficient of $Q^C \Delta^J S^K$ in the Nth exponential Taylor coefficient of the corrected curvature.

There is a useful probabilistic simplification of (74). Let

$$T(C, J, K) = \sum_{c,j,k} W(c, j, k), \qquad \pi(c, j, k) = \frac{W(c, j, k)}{T(C, J, K)},$$

with W as in (78) below, and regard $n = 3c + 2j + k$ as a random variable under π. Complementing the cut sends n to $N - n$ without changing its weight, so $\mathbb{E}_\pi n = N/2$. Since the kernel in (74) is

$$2 - \frac{N}{2} + 2\left(n - \frac{N}{2}\right)^2,$$

one obtains the exact identity

$$\Gamma(C, J, K) = T(C, J, K)\left(2 - \frac{N}{2} + 2\,\mathrm{Var}_\pi(n)\right). \tag{75}$$

Thus the uniform coefficient sign assertion is equivalent to the elementary cut-dispersion bound

$$\mathrm{Var}_\pi(n) \geq \frac{N}{4} - 1. \tag{76}$$

The five exceptional triples below are exactly the cases in which this conditional cut distribution is slightly too concentrated. This variance formulation is also compatible with conditioning first on the number c of cycles to the left of the cut, so the law of total variance offers a second route to the desired finite-exception statement.

Exact expansion suggests the following particularly rigid sign pattern:

$$\Gamma(C, J, K) \geq 0$$

except when $(C, J) = (2, 0)$ and $3 \leq K \leq 7$. This has been checked in exact integer arithmetic for $1 \leq C \leq 12$, $0 \leq J \leq 9$, and $0 \leq K \leq 20$; it is not yet a proof for all indices. The five exceptions do not threaten the desired curvature. They occur at total degrees $N = 9, \ldots, 13$ and combine

with the neighboring $C = 1$ and $C = 3$ terms into positive quadratics. After discarding other nonnegative terms, the relevant brackets are

N	quadratic in $x = Q/S^3$
9	$133 - 320x + 6592x^2$
10	$47 - 437x + 18096x^2$
11	$171 - 3280x + 309952x^2$
12	$85 - 2296x + 569728x^2$
13	$55 - 920x + 1179744x^2$.

Their discriminants are respectively

$$-3404544, \quad -3211079, \quad -201248768, \quad -188435904, \quad -258697280,$$

so all five are strictly positive. Consequently a proof of the uniform sign assertion for (74) would settle the first canonical start coefficient by coefficient. Analogous expansions for the other two starts show no exceptional negative group in the tested range. This finite-exception binomial lemma is now the most concrete algebraic target alongside the Riccati inequalities.

There is evidence for a sharper reduction of that lemma. Normalize by the positive coefficient in (73) and set

$$R(C, J, K) = \frac{\Gamma(C, J, K)}{\alpha(C, J, K)}.$$

Exact arithmetic through $C \le 9$, $J \le 14$, and $K \le 20$ gives

$$\begin{aligned} &R(C, J+1, K) \ge R(C, J, K), && C \ge 1, \\ &R(C, J, K+1) \ge R(C, J, K), && (C, J, K) \notin \{(2, 0, k) : 0 \le k \le 5\}. \end{aligned}$$

If these two monotonicity statements hold in general, the whole uniform sign claim reduces to the already proved equal-rate value

$$R(C, 0, 0) = \frac{2^{3C+1} + (-1)^C(-27C^2 + 9C + 4)}{3}$$

and the single slice $C = 2$, $J = 0$. That slice has the closed form

$$\begin{aligned} \Gamma(2, 0, K) = \frac{1}{6\,2^K}\big\{&96\,2^K(K^2 - 5K - 34) \\ &+ K^5 + 23K^4 + 219K^3 + 1081K^2 + 2888K + 3348\big\}. \end{aligned} \tag{77}$$

Formula (77) follows by writing the interior summation with $n = k + 3$, extending it to the full binomial sum $0 \le n \le K + 6$, evaluating the moments through degree four, and subtracting the six boundary terms. It is positive for $K \ge 9$ directly, because both displayed groups are then nonnegative, and inspection of $0 \le K \le 8$ leaves exactly $K = 3, \dots, 7$ negative. Thus the two normalized monotonicity inequalities, rather than an unrestricted three-variable sign calculation, are sufficient for the first start.

There is an exact insertion formula for the second monotonicity statement. It substantially narrows what remains to be proved. Put

$$\kappa_N(n) = 2 - \frac{N}{2} + \frac{(N - 2n)^2}{2}, \qquad W(c, j, k) = \binom{N}{n}\alpha(c, j, k)\alpha(C - c, J - j, K - k),$$

where $n = 3c + 2j + k$ and $N = 3C + 2J + K$. Inserting one S-loop on either side of the cut gives

$$R(C, J, K+1) - R(C, J, K) = \frac{1}{\alpha(C, J, K)} \sum_{c,j,k} W(c, j, k) \Phi_{N,C}(n, c), \tag{78}$$

where

$$\Phi_{N,C}(n, c) = \frac{N+1}{N-C} \left\{ \frac{n-c}{n+1} \kappa_{N+1}(n+1) + \frac{N-n-C+c}{N+1-n} \kappa_{N+1}(n) \right\} - \kappa_N(n).$$

The identity uses only

$$(K+1)\alpha(C, J, K+1) = \frac{N-C}{2} \alpha(C, J, K)$$

and its two cut analogues. In the centered variables $u = N - 2n$ and $r = C - 2c$, the kernel becomes

$$\Phi_{N,C}(n, c) = \frac{A_N(u^2)}{2((N+2)^2 - u^2)} + \frac{(N+1)(3N - u^2)u(u-r)}{(N-C)((N+2)^2 - u^2)}, \tag{79}$$

with

$$A_N(v) = v^2 + (N^2 - 3N)v - N^3 + 2N^2 + 8N.$$

Both W and Φ are invariant under the complementary-cut involution $(n, c) \mapsto (N - n, C - c)$. The kernel can be negative, so pointwise insertion does not prove monotonicity. Equation (78) shows that the missing statement is precisely a correlation estimate between the total cut imbalance u and its noncycle part $u - r$; it replaces a comparison of two triple sums by one weighted average over the old cuts.

The first normalized monotonicity has a parallel exact insertion formula. It is useful because it puts both conjectural comparisons on the same old-cut probability space. With N, n, W as above, set

$$m = n - c, \qquad \bar{m} = N - n - C + c,$$

and define

$$\begin{aligned}
\lambda_L &= \frac{(N+2)(N+1)m(m+1)}{8(n+2)(n+1)(2c+2j+1)}, \\
\lambda_R &= \frac{(N+2)(N+1)\bar{m}(\bar{m}+1)}{8(N-n+2)(N-n+1)(2(C-c)+2(J-j)+1)}, \\
\Psi_{C,J,K}(c, j, k) &= \frac{8(2C+2J+1)}{(N-C)(N-C+1)} \left\{ \lambda_L \kappa_{N+2}(n+2) + \lambda_R \kappa_{N+2}(n) \right\} - \kappa_N(n).
\end{aligned}$$

Then

$$R(C, J+1, K) - R(C, J, K) = \frac{1}{\alpha(C, J, K)} \sum_{c,j,k} W(c, j, k) \Psi_{C,J,K}(c, j, k). \tag{80}$$

Indeed,

$$\frac{\alpha(c, j+1, k)}{\alpha(c, j, k)} = \frac{(n-c)(n-c+1)}{8(j+1)(2c+2j+1)},$$

and the same identity on the complementary side accounts for every newly inserted pair exactly $J + 1$ times. Formula (80) follows after using the corresponding total ratio and the two binomial-coefficient ratios. Like the loop kernel, Ψ can be negative pointwise; the assertion needed is its nonnegative weighted mean.

There is also a positive beta-integral form which eliminates the loop split k from either weighted average. Put

$$\beta(C,J,K)=\frac{\alpha(C,J,K)}{(3C+2J+K)!}.$$

For $C\ge 1$, a beta integral gives

$$\beta(C,J,K)=\frac{d_{C,J}}{2^K K!}\int_0^1 x^{2C+2J+K-1}(1-x)^C\,dx,\qquad d_{C,J}=\frac{(C)_J}{4^J J!(2C+2J-1)!C!}. \tag{81}$$

For an interior split $1\le c\le C-1$, fix j and introduce independent integration variables x,y from the two copies of (81). If

$$p=\frac{x}{x+y},\qquad d=3c+2j,$$

then summing all $k=0,\dots,K$ replaces the loop allocation by $X\sim\mathrm{Bin}(K,p)$ and gives

$$\begin{aligned}\sum_{k=0}^{K}&\kappa_N(d+k)\beta(c,j,k)\beta(C-c,J-j,K-k)\\ &=\frac{d_{c,j}d_{C-c,J-j}}{2^K K!}\int_0^1\int_0^1 x^{2c+2j-1}(1-x)^c y^{2(C-c)+2(J-j)-1}(1-y)^{C-c}\\ &\qquad\times(x+y)^K\,\Xi_{N,K,d}(p)\,dx\,dy,\end{aligned} \tag{82}$$

where the remaining kernel is merely the quadratic

$$\Xi_{N,K,d}(p)=2-\frac{N}{2}+2\left(d+Kp-\frac{N}{2}\right)^2+2Kp(1-p). \tag{83}$$

The two boundary cuts contribute $2\kappa_N(0)\beta(C,J,K)$. Since $\Gamma/N!=\sum\kappa_N(n)\beta(c,j,k)\beta(C-c,J-j,K-k)$, equations (81)–(83) are an exact double-integral formula for $R(C,J,K)$. Thus the k-sum and its changing support are not intrinsic: after conditioning on x,y, loop allocation is an ordinary binomial random variable. What remains is to control the negative constant in (83) by the square, binomial variance, and the two positive boundary cuts. This form uses the actual α-weights and therefore avoids the false generic renewal-point variance statement, which ignores the averaging over loop allocations.

For exact computation there is an even simpler coefficient system which keeps the rate ordering explicit. Put $d=b-a\ge 0$. Then the coefficient of

$$Q^C a^J d^K z^{3C+J+K}$$

in the first-start return series is

$$\widetilde{\alpha}(C,J,K)=\binom{C+K-1}{K}\binom{2C+K+J-1}{J}\qquad(C\ge 1). \tag{84}$$

To see this, expand $(1-az)^{-C}(1-(a+d)z)^{-C}$ and apply Vandermonde's identity. The analogous coefficients for the second and third starts are respectively

$$\binom{C+K-1}{K}\binom{2C+K+J}{J},\qquad\binom{C+K}{K}\binom{2C+K+J+1}{J},$$

with the evident convention at $C=0$. In exact tests through total Taylor degree 20, the corrected-curvature coefficients for starts two and three are all nonnegative. For start one the only negative

monomials again have $C = 2$ and total Taylor degree 9 through 13: in the ordered coordinates they are the two families $Q^2a^Ld^0$ and $Q^2a^{L-1}d$, $3 \le L \le 7$. Formula (84) is often more convenient than (73) for a direct Vandermonde proof; the symmetric coordinates remain better for combining the exceptional terms into the five quadratics above.

The boundary $a = 0$ is especially simple in these coordinates. Write $\widetilde{\Gamma}(C, K)$ for the coefficient of $Q^C d^K$ in the corrected first-start curvature and put

$$\widetilde{R}(C, K) = \frac{\widetilde{\Gamma}(C, K)}{\widetilde{\alpha}(C, 0, K)}, \qquad N = 3C + K.$$

There is an exact loop-insertion identity parallel to (78). If

$$\widetilde{W}(c, k) = \binom{N}{3c + k}\widetilde{\alpha}(c, 0, k)\widetilde{\alpha}(C - c, 0, K - k), \qquad n = 3c + k,$$

then

$$\widetilde{R}(C, K + 1) - \widetilde{R}(C, K) = \frac{1}{\widetilde{\alpha}(C, 0, K)} \sum_{c,k} \widetilde{W}(c, k)\widetilde{\Phi}_{C,K}(c, k), \tag{85}$$

where

$$\widetilde{\Phi}_{C,K}(c, k) = \frac{N + 1}{C + K}\left\{\frac{c + k}{n + 1}\kappa_{N+1}(n + 1) + \frac{C - c + K - k}{N + 1 - n}\kappa_{N+1}(n)\right\} - \kappa_N(n).$$

Indeed, mark and remove one of the $K + 1$ loops in the new coefficient, on either side of the cut, and use

$$(K + 1)\widetilde{\alpha}(C, 0, K + 1) = (C + K)\widetilde{\alpha}(C, 0, K).$$

This accounts for every new term exactly $K + 1$ times and proves (85).

The initial value $\widetilde{R}(C, 0)$ is the proved equal-rate expression in (12). Exact arithmetic finds

$$\widetilde{R}(C, K + 1) \ge \widetilde{R}(C, K)$$

for $1 \le C \le 12$ and $0 \le K \le 20$, with the sole exception $(C, K) = (2, 0)$. That exception is harmless, and in fact the entire $C = 2$ slice can be evaluated explicitly:

$$\widetilde{\Gamma}(2, K) = 128\,2^K - \frac{K^4 + 14K^3 + 69K^2 + 156K + 228}{2} > 0. \tag{86}$$

To verify the strict inequality, check $K = 0, 1, 2$ directly. Thereafter it propagates by induction, since for $P(K) = K^4 + 14K^3 + 69K^2 + 156K + 228$ one has

$$2P(K) - P(K + 1) = K^4 + 10K^3 + 21K^2 - 28K - 12 > 0 \qquad (K \ge 2).$$

The $C = 1$ slice is even simpler: $\widetilde{\Gamma}(1, K) = K^2 + 5K + 10$. Consequently, proving that the weighted mean in (85) is nonnegative for $C \ge 3$ would give an independent coefficientwise proof of the $a = 0$ boundary already established in Proposition 18. The kernel itself changes sign, so this assertion is again a correlation inequality rather than a termwise estimate. All identities and the quoted finite check are verified in `python/check_canonical_k3_coefficients.py`.

6 Equivalent walk-count formulation

The Taylor expansion

$$F(t) = \sum_{n \geq 0} \frac{t^n}{n!} u^T A^n v$$

has nonnegative coefficients. Define a probability distribution by

$$\Pr_t(N = n) = \frac{t^n u^T A^n v}{n! F(t)}.$$

Then

$$\mathbb{E}_t N = t\frac{F'(t)}{F(t)}, \qquad \mathrm{Var}_t(N) = t\frac{F'(t)}{F(t)} + t^2 (\log F(t))''.$$

Consequently (5) is exactly

$$\mathrm{Var}_t(N) \geq \mathbb{E}_t N - (s-1). \tag{87}$$

Here $u^T A^n v$ is the total weight of length-n walks in a weighted directed graph with s vertices. Thus the matrix lemma says that conditioning a Poisson walk by its endpoints can reduce its variance below its mean by at most $s-1$. A likely combinatorial proof should remove loops from a walk: at most $s-1$ steps survive loop erasure, while all remaining steps must carry the Poisson-type variance on the right side of (87).

It is tempting to seek a coefficientwise strengthening of this formulation. Write

$$D(t) = t^2\big(F(t)F''(t) - F'(t)^2\big) + (s-1)F(t)^2 = \sum_{N \geq 0} D_N \frac{t^N}{N!},$$

and put $a_n = u^T A^n v$. Direct expansion and symmetrization give

$$D_N = \sum_{i=0}^{N} \binom{N}{i} \left(s - 1 - \frac{N}{2} + \frac{(N-2i)^2}{2} \right) a_i a_{N-i}. \tag{88}$$

If I has the symmetric distribution

$$\Pr(I = i) = \frac{\binom{N}{i} a_i a_{N-i}}{\sum_{j=0}^{N} \binom{N}{j} a_j a_{N-j}},$$

then $D_N \geq 0$ is precisely

$$\mathrm{Var}(I) \geq \frac{N}{4} - \frac{s-1}{2}. \tag{89}$$

The normalizing denominator counts two weighted walks of total length N, together with an interleaving of their steps. Thus I is the length of the first walk. However, the proposed coefficientwise inequality is false, even though all numerical tests of the pointwise sum $D(t)$ remain nonnegative. Take $s = 4$, let A be the adjacency matrix of the directed four-cycle (all four edge weights equal to one), and take $u = v = e_1$. Then

$$a_n = \begin{cases} 1, & 4 \mid n, \\ 0, & 4 \nmid n. \end{cases}$$

For $N = 8$, only $i = 0, 4, 8$ contribute to (88), and hence

$$D_8 = 31 + 31 - \binom{8}{4} = -8 < 0.$$

Equivalently, I is supported on $0, 4, 8$, with weights $1, 70, 1$, so

$$\mathrm{Var}(I) = \frac{4}{9} < \frac{1}{2} = \frac{N}{4} - \frac{s-1}{2}.$$

Thus cancellation between different Taylor coefficients is essential; a coefficient-by-coefficient tail-switching proof cannot establish Conjecture 2. Notice that the same example gives

$$F(t) = \frac{\cosh t + \cos t}{2},$$

and it does not furnish a pointwise counterexample to (5).

7 A state-return route beyond three moments

There is a second possible route to the all-state conjecture which does not pass through the matrix-exponential statement. Remove one transient state and decompose the absorption time into a geometric number of return excursions followed by one terminal excursion. If x is the mean holding time at the removed state and r is the mean number of returns, the moment generating functions satisfy

$$M_Z(z) = \frac{V(z)}{(1+r)(1-xz) - rU(z)}.$$

This suggests the following uniform induction lemma: if both excursion laws, of order at most $n-1$, satisfy the hierarchy through level k, then the renewal law satisfies level k at order n. Such a lemma for every k would prove the full conjecture by induction on the number of states. It is already known at $k = 1, 2, 3$.

The first unproved instance, $k = 4$, now has a reasonably small exact reduction. Write the first five return-excursion moments as

$$(u, A, C, E, G)$$

and the terminal-excursion moments as (v, B, D, F, J). After adjoining the exponential holding time, let a_j, b_j be the return and terminal kernel moments. Then

$$M_j = b_j + ra_j + \sum_{\ell=1}^{j-1} \binom{j}{\ell} ra_\ell M_{j-\ell}.$$

The sharp fifth-moment child bounds permit the substitutions

$$G = \frac{n+3}{n+2}\frac{E^2}{C}, \qquad J = \frac{n+3}{n+2}\frac{F^2}{D}.$$

The resulting parent gap

$$\mathcal{G}_4 = (n+3)M_3M_5 - (n+4)M_4^2$$

is a strictly convex quadratic in E, F. If

$$R_3 = M_3 - rC - D,$$

its Hessian has off-diagonal entry $-2r(n+4)$, and its determinant is

$$\frac{4r(n+3)^2M_3}{CD(n+2)^2}\left(rC + D + (n+3)^2R_3\right) > 0.$$

Thus the top-moment convexity used in the $k = 3$ proof persists at $k = 4$.

The stationary-point algebra simplifies substantially after a weighted change of variables. Put

$$\alpha = \frac{n+3}{n+2}, \quad c = rC, \quad d = D, \quad w = c + d, \quad y = \frac{E}{C}, \quad f = \frac{F}{D},$$

and set

$$m = M_1, \qquad P = M_3, \qquad T = x + ra_1,$$
$$L = M_4\big|_{E=F=0}, \qquad K = M_5\big|_{E=F=0}.$$

Then the entire quadratic is

$$\begin{aligned}\mathcal{G}_4 = (n+3)P\{K + \alpha(cy^2 + df^2) + 5c(T+m)y + 5dTf\} \\ - (n+4)(L + cy + df)^2.\end{aligned} \tag{90}$$

Introduce the weighted mean and difference

$$\mu = \frac{cy + df}{w}, \qquad \delta = f - y.$$

The two variables separate in (90). In particular, every interior stationary point satisfies the simple identity

$$\delta_* = \frac{5m}{2\alpha} = \frac{5(n+2)}{2(n+3)}M_1. \tag{91}$$

After minimizing in δ, the remaining quadratic in μ has leading coefficient

$$A_\mu = \frac{w}{n+2}\Delta, \qquad \Delta = (n+3)^2(P - w) + w > 0,$$

linear coefficient

$$B_\mu = 5(n+3)P(wT + cm) - 2(n+4)Lw,$$

and constant term

$$C_\mu = (n+3)P\left(K - \frac{25cdm^2}{4\alpha w}\right) - (n+4)L^2.$$

Consequently the value at the unconstrained interior stationary point is

$$C_\mu - \frac{B_\mu^2}{4A_\mu}. \tag{92}$$

The lower fourth-moment child bounds say

$$y \geq \frac{n+2}{n+1}\frac{C}{A}, \qquad f \geq \frac{n+2}{n+1}\frac{D}{B}.$$

The algebraic lemma established below is that (92) is nonnegative whenever the stationary point given by (91) satisfies these two feasibility inequalities and the lower child hierarchy holds.

The quantities in this last lemma also obey a short recurrence. With $a = rA$ and $b = B$,

$$\begin{aligned}M_2 &= 2Tm + a + b, \\ P &= 3TM_2 + 3am + c + d, \\ L &= 4TP + 6aM_2 + 4cm, \\ K &= 5TL + 10aP + 10cM_2.\end{aligned}$$

These identities remove all large kernel expansions from the interior problem. Random searches over the child moment cone have found feasible interior stationary points, but none with a negative value in (92).

There is a sharper reduction of this last problem. By homogeneity put $m = 1$, write $f = y + 5(n+2)/(2(n+3))$, and let $p, q, A_0, B_0 \in [0,1]$. The feasibility inequalities and the two third-moment child inequalities are parametrized by

$$a = \frac{nTypA_0}{n+2}, \qquad c = \frac{ayp}{\beta}, \qquad b = \frac{n(1-T)fqB_0}{n+2}, \qquad d = \frac{bfq}{\beta}, \qquad \beta = \frac{n+2}{n+1}.$$

Indeed $p, q \le 1$ are exactly the two stationary-point feasibility bounds, while $A_0, B_0 \le 1$ encode $nTc \ge (n+1)a^2$ and $n(1-T)d \ge (n+1)b^2$. The remaining terminal second-moment bound is

$$\frac{(n-1)B_0qf}{n+2} \ge 1 - T.$$

Define

$$\begin{aligned} S &= 2(n+4)M_4 - (n+3)P(2\alpha f + 5T), \\ R &= M_5 - M_4\left(\alpha f + \frac{5}{2}T\right). \end{aligned}$$

Then the parent gap has the exact identity

$$\mathcal{G}_4 = (n+3)PR - \frac{1}{2}M_4S.$$

Both S and R are affine in B_0. If

$$S = S_0 + B_0S_1, \qquad R = R_0 + B_0R_1, \qquad J = R_0S_1 - R_1S_0,$$

then at an interior stationary point $S = 0$,

$$R = \frac{J}{S_1}.$$

On finite positive parameters, the exact certificates below prove

$$S(1) \ge 0 \quad \Longrightarrow \quad S_1 > 0,\ J \ge 0.$$

The strict slope and the orientation argument below justify both division by S_1 and the condition $S(1) \ge 0$ at a stationary point. The terminal second-moment bound is unnecessary for this interior step.

After compactifying $n - 4$ and y, the sole common infinity corner of $S(1), S_1, J$ has Newton boundary

$$(0,1), \qquad (2,0).$$

Thus it is covered by the two charts associated with $1/y \asymp 1/n^2$. Their radial faces reduce the remaining implication to five-variable polynomials with respectively $14, 4, 20$ terms. On one secondary face the slope implication follows directly from convexity in A_0: if

$$H = 12A_0p - 2\theta q + 3q - 6,$$

then the condition polynomial is convex in A_0, is nonpositive at $A_0 = 0$, and at the root of H contains the nonnegative factor

$$6 - 3p + T(3 - 2\theta)(q - p).$$

Hence the condition being nonnegative forces $H \geq 0$.

The analogous slope implication on the opposite infinity face is now also exact. Write $F = S(1)/4$, $H = S_1/2$, and put

$$K_0 = 3TWq - 6TW - 2q, \qquad L = 3TWp - 3TWq + 3Wp - 6W - 2p + 2q.$$

Then $H = 12A_0TWp + K_0$, $F(0) = -q(T-1)K_0 \leq 0$, and at the root of H,

$$F = -\frac{K_0 L}{12W}.$$

Convexity in A_0 therefore leaves only endpoint estimates. With $C = 3W(T+1) - 2$ and $D = 2 - 3TW$, the constraint $H(1) \geq 0$ gives $p \geq \frac{1}{2} + Dq/(12TW)$. If $C \geq 0$, maximizing $L = Cp + Dq - 6W$ at the appropriate endpoints gives either $-3W$ or $3W(T-1) - 2$. If $C \leq 0$, substitution of this lower bound leaves $-Q/(12TW)$; Q is affine in q, its $q = 0$ endpoint is manifestly nonnegative, and its $q = 1$ endpoint has nonnegative Bernstein coefficients after $W = 2z/[3(T+1)]$ and one split at $z = 1/2$. For the remaining endpoint put $E = 15TW + 3W - 2$. When $E \leq 0$, the polynomial $-3(5T+1)F(1)$, after $W = 2z/[3(5T+1)]$, has nonnegative Bernstein coefficients. When $E \geq 0$, convexity in p reduces to the preceding root or to $p = 1$; in the latter case $H \leq 0$ gives $q \geq 6TW/(2 - 3TW)$, and $F(1)$ is decreasing in q and already nonpositive at that lower endpoint. Thus $S(1) \geq 0$ implies $S_1 \geq 0$ on both infinity faces.

One resultant face is exact as well. On the $T \leq Z$ secondary chart (which is the same polynomial chart as the opposite primary face), put $K = J/(8A_0p)$. An exact rational identity has been found of the form

$$K = MF + NH + P,$$

where every Bernstein coefficient of M, N, P is nonnegative. Hence $F \geq 0$ implies $J \geq 0$ there.

On the remaining $Z \leq T$ chart, both F and $K = J/(8A_0p)$ are affine in T. The endpoint $T = 0$ now has another exact identity $K = MF + NH + P$ with nonnegative rational Bernstein coefficients. At $T = 1$, put $x = A_0p$. Then

$$F = 2x\{6x + (3 - R)p - 6\}.$$

At its boundary $x_0 = 1 - (3 - R)p/6$, the target factors as

$$K(x_0) = \frac{1}{3}\{2Rp - 2Rq - 6p + 3q + 6\}\{10Rp^2 - 3Rp - 30p^2 + 60p - 18\}.$$

Feasibility gives $p \geq 6/(9 - R)$, which makes both factors nonnegative. If $p \geq q$, the target is convex in x and its derivative at x_0 is nonnegative; if $p \leq q$, it is concave and its second endpoint $x = p$ is nonnegative by checking the two affine endpoints $q = p, 1$. Thus both T-endpoints are exact. The interior crossing $F(T) = 0$ is now exact as well. Its unresolved boxes converge to the single equality corner

$$T = 1, \qquad R = 0, \qquad p = 1, \qquad q = 0, \qquad x = A_0p = \frac{1}{2}, \qquad H = 0.$$

Put $u = 1 - T$, $a = 1 - p$, and $h = H$, and use the weighted variables

$$h, \qquad 6a, \qquad 3q, \qquad 2R, \qquad 3u.$$

In the chart where h dominates, the exact Bernstein coefficients of K/h all lie between 2 and 10. In each of the other four charts, all Bernstein coefficients of $-F$ are nonnegative. This closes the corner. The complement of the local box is partitioned at

$$u = \frac{1}{3}, \quad R = \frac{1}{2}, \quad a = \frac{1}{6}, \quad q = \frac{1}{3}, \quad h = 1.$$

Twenty-nine of its rational boxes close by exact conditional subdivision, at depth at most six. One exceptional box has $-F \geq 0$, with its sole zero on the weighted fan boundary. In the other, feasibility can be an equality only at $R = 0, a = 1/6, q = 1/3, h = 5$; there

$$F = -\frac{5}{4}(3u - 2), \qquad K = \frac{694 - 387u}{36},$$

so $F \geq 0$ again gives $K > 0$. Consequently the full infinity-face implication $S(1) \geq 0 \Rightarrow S_1, J \geq 0$ is exact. The transverse problem can now be sharpened considerably. Write $F = S(1)$ and $H = S_1$. In the $U^2 \leq V$ chart, $H = h_0 + h_1 A_0$ is affine in A_0, with

$$h_1 = -24Tp(D-1)(D+1)(3DW+1)(5DW+1)(6DW+1)^2(7DW+1) \geq 0,$$

whereas $F = f_0 + f_1 A_0 + f_2 A_0^2$ is convex, since

$$\frac{\partial^2 F}{\partial A_0^2} = 96T^2p^2(D-1)^2(D+1)^2(3DW+1)^2(5DW+1)(6DW+1)^3(7DW+1) \geq 0.$$

At the root $A_* = -h_0/h_1$, clear the denominator and put

$$\Phi = h_1^2 F(A_*) = f_2 h_0^2 - f_1 h_0 h_1 + f_0 h_1^2.$$

The two ordinary blow-up charts

$$D = R, \quad W = Rx, \qquad \text{and} \qquad D = Rx, \quad W = R$$

have only nonnegative Bernstein coefficients in $-\Phi$. The arrays have shapes $43 \times 18 \times 6 \times 4 \times 3$ and $43 \times 26 \times 6 \times 4 \times 3$. An exact zero-face audit shows $\Phi < 0$ whenever $0 < D < 1$, $W > 0$, $T, p > 0$, including $q = 0$. In the first blow-up the maximal zero faces are

$$R = 0, T = p = 1, q = 0; \qquad R = 1; \qquad T = 0; \qquad p = 0.$$

In the second, replace $R = 1$ by $R = x = 1$. Every one is excluded by these finite positive parameters. Direct Bernstein checking also gives $F(0) < 0$ on the same domain: its maximal zero faces are

$$D = 0, T = 1; \quad D = q = 0; \quad W = T = 0; \quad T = q = 0.$$

Since $h_1 > 0$, convexity between $A_0 = 0$ and the root $A_* = -h_0/h_1$ therefore proves the strict implication

$$F \geq 0 \quad \Longrightarrow \quad H > 0.$$

This statement includes $q = 0$ and $q = 1$; it is asserted on finite positive parameters rather than every degenerate compactification face. The full Jacobian certificate for this chart is given below.

For completeness, the Newton analysis leading to the remaining J charts is as follows. The first singular stratum is

$$W = T = 0, \qquad p = q = 1, \qquad A_0 = \frac{1}{3},$$

with $D \in [0,1]$ free. For $P = 1-p$, $Q = 1-q$, and $C = A_0 - 1/3$, its first normal terms are

$$F = -5D^2(7D^2+8)T + O_2,$$
$$H = -(17D^2-2)T + O_2,$$
$$J = \frac{1}{3}T(D^2-1)(3D^2+2)(258D^4 - D^2 - 32) + O_2.$$

Use the six weighted variables

$$D^2, \qquad W^2, \qquad T, \qquad 1-p, \qquad 1-q, \qquad |A_0 - 1/3|.$$

All six lower-A_0 charts and the upper D^2-dominant chart have $-F \geq 0$ coefficientwise. In each of the three singular upper charts where P, Q, or C is dominant, put $W = R^2 v$. At the repeated $t = v = 0$ face the tangents have the form

$$F = at - bv^2, \qquad H = ht - ev^2, \qquad J = jt + gv^2, \qquad b, e, g \geq 0.$$

The slope determinant $hb - ea$ factors with a nonnegative sign on the unit cube. For the Jacobian, $a > 0$ implies $j \geq 0$ by exact conditional dyadic certificates: four cuts in the P chart, four in the Q chart, and a depth-twelve tree in the C chart with at most nine unresolved boxes at any level. Hence all three repeated equality faces themselves are exact.

The full P-dominant radial extension is now exact as well. In its secondary t-dominant chart use

$$D = rd, \quad W = r^2\sigma u, \quad T = r^2\sigma^2, \quad p = 1 - r^2, \quad q = 1 - r^2\bar{q}, \quad A_0 = \frac{1}{3} + \frac{2}{3}r^2 c,$$

and resolve the remaining common corner by $r = R^2\rho$, $\sigma = R$. After removing positive monomial factors, the condition polynomial F has multidegree $(75, 11, 2, 2, 33, 7)$ and 799 terms, while J has multidegree $(106, 16, 1, 3, 48, 8)$ and 4199 terms. A guided midpoint tree closes at depth eleven with fourteen leaves: six have $J \geq 0$ coefficientwise and eight have $-F \geq 0$ coefficientwise. This is an exact integral Bernstein computation, not a floating diagnostic. To keep memory bounded, the last variable is streamed in eight F slices and nine J slices; the remaining slice shapes are respectively $76 \times 12 \times 3 \times 3 \times 34$ and $107 \times 17 \times 2 \times 4 \times 49$. Dyadic subdivision is replayed by integral de Casteljau additions, so no dense six-dimensional rational array is formed.

The analogous C-dominant radial chart is now exact. Keep the same D, W, T substitutions, but put

$$p = 1 - r^2\bar{p}, \qquad q = 1 - r^2\bar{q}, \qquad A_0 = \frac{1}{3} + \frac{2}{3}r^2, \qquad r = R^2\rho, \quad \sigma = R.$$

In the order $(R, d, \bar{p}, \bar{q}, \rho, u)$, the condition again has multidegree $(75, 11, 2, 2, 33, 7)$ and 799 terms. The Jacobian has multidegree $(106, 16, 4, 1, 48, 8)$ and 4199 terms. Its exact midpoint certificate has 75 leaves at depth at most 19: 44 certify $J \geq 0$ and 31 certify $-F \geq 0$. The terminal ratio u needs only the split at $1/2$. Leaves with no u cut are checked on the original interval; the others are checked after the appropriate half-interval substitution. Each restricted polynomial is streamed in its terminal Bernstein slices, and only five-dimensional integer arrays are subdivided.

For the corresponding Q chart put

$$p = 1 - r^2\bar{p}, \qquad q = 1 - r^2, \qquad A_0 = \frac{1}{3} + \frac{2}{3}r^2 c, \qquad r = R^2\rho, \quad \sigma = R.$$

The condition has the same multidegree and term count, while J has multidegree $(106, 16, 4, 3, 48, 8)$ and 4199 terms in $(R, d, \bar{p}, c, \rho, u)$. Direct subdivision reveals an additional equality corner

$$R = \rho = 1, \qquad d = u = \bar{p} = 0, \qquad c = \frac{1}{4}.$$

It is resolved using the exact identity

$$K = J - 4F, \qquad M = -\left.\frac{\partial^2 K}{\partial d\,\partial u}\right|_{R=\rho=1,\ d=u=0}, \qquad J = 4F + M\left(d - \frac{u}{2}\right)^2 + E.$$

Here M depends only on $(\bar{p}, c)$ and has 20 terms; E has 4770 terms and the same multidegree as J. The squared term removes the adverse mixed tangent. At the equality corner,

$$M = 64, \qquad K = 480(1-R) + 192(1-\rho) + 96\bar{p} + \frac{640}{3}\left(c - \frac{1}{4}\right)^2 + 96d^2 - 64du + 32u^2 + \cdots.$$

The exact midpoint certificate has 38 leaves at depth at most 35: 16 use $J \geq 0$, 11 use $K \geq 0$, 9 use $-F \geq 0$, and 2 use both $M \geq 0$ and $E \geq 0$. All integral Bernstein slices pass. Thus both the Q- and C-dominant full radial implications are exact. For a condition-negative leaf, the checker also enumerates all cube faces on which every active Bernstein coefficient vanishes. The Jacobian restricted to each such face is exactly constant and nonnegative (in fact zero except at one C-chart point, where it is $64/3$). Consequently the certificates include the equality case $F = 0$ as well as $F > 0$.

The full upper W-dominant chart is exact as well. Use

$$D = rd, \quad W = r, \quad T = r^2 t,$$
$$p = 1 - r^2\bar{p}, \quad q = 1 - r^2\bar{q}, \quad A_0 = \frac{1}{3} + \frac{2}{3}r^2 c.$$

After division by r^2, their multidegrees and support sizes are

	multidegree	terms
F	$(28, 11, 3, 2, 2, 2)$	799
J	$(40, 16, 3, 4, 1, 3)$	4199

Their radial faces are -8 and $8(8t+3)/3$. In particular the condition is strictly negative on $0 \leq r \leq 1/2$. The full chart has an exact midpoint certificate with 131 leaves at depth at most 23: 70 use $-F \geq 0$ and 61 use $J \geq 0$. All condition equality faces pass the same exact Jacobian audit.

The opposite infinity chart $V \leq U^2$ now also has an exact full radial Jacobian certificate. In the original coordinates (U, W, T, p, q, A_0), F has multidegree $(11, 3, 3, 2, 2, 2)$ and 356 terms; J has multidegree $(16, 5, 3, 4, 1, 3)$ and 1094 terms. Its complete midpoint tree has 393 leaves at depth at most 25, comprising 218 condition-negative leaves and 175 Jacobian-positive leaves. All slices and condition equality faces have passed exact verification.

The slope implication on this opposite chart is exact for admissible finite interior parameters. Write $H = h_0 + h_1 A_0$ and $F = f_0 + f_1 A_0 + f_2 A_0^2$. Then

$$h_1 = -24TWp(3U+1)(5U+1)(6U+1)^2(7U+1)(U^2W-1),$$
$$2f_2 = 96T^2Wp^2(3U+1)^2(5U+1)(6U+1)^3(7U+1)(U^2W-1)^2.$$

Both have the required nonnegative sign. The endpoint $-F(0)$ has nonnegative Bernstein coefficients. At the $H = 0$ root, the cleared value factors as

$$-h_1^2 F(-h_0/h_1) = -48T^2Wp^2(3U+1)^2(5U+1)(6U+1)^3(7U+1)(U^2W-1)^2 G,$$

where G is affine in p. Its $p = 1$ endpoint satisfies $G \leq 0$ coefficientwise. At the feasible lower endpoint $p = -h_0/(h_1/p)$, the cleared value reduces to a four-variable polynomial. Blow up $(T, q) =$

$(0,0)$ by $q = Tz$ and $T = qz$. The resulting exact root trees have 9 and 47 leaves, respectively, at depths at most 8 and 11. The $A_0 = 1$ endpoint has exact conditional trees with 991 and 1731 leaves, at depths at most 27 and 23. All their zero faces lie on excluded degenerate parameter faces; the root value is strictly negative there, and convexity proves $F \geq 0 \Rightarrow H > 0$. The $G(p = 1)$ zero faces are only $W = 0, q = 0$, $W = 0, q = 1$, and $T = q = 0$. The $F(0)$ and root-tree zero-face audits likewise exclude no face defined by $q = 0$ alone. Thus the strict implication includes $q = 0$ as needed for stationary orientation. This statement does not assert the stronger implication on every face of the closed compactification.

The full upper T-dominant chart is now exact too. Set

$$D = rd, \quad W = rw, \quad T = r^2,$$
$$p = 1 - r^2\bar{p}, \quad q = 1 - r^2\bar{q}, \quad A_0 = \frac{1}{3} + \frac{2}{3}r^2c.$$

After division by r^2, F and J have multidegrees $(28, 11, 7, 2, 2, 2)$ and $(40, 16, 8, 4, 1, 3)$, with 799 and 4199 terms. The equality corner is $r = 1, d = w = \bar{p} = 0, \bar{q} = 1, c = 1/4$. Using the same 20-term coefficient $M(\bar{p}, c)$ as in the Q certificate gives

$$J = 4F + M\left(d - \frac{w}{2}\right)^2 + E.$$

The exact tree has 49 leaves at depth at most 31: 15 use J, 17 use $J - 4F$, 16 use $-F$, and one uses both M and E. All slices and equality faces pass, including one nonconstant face on which $J = 48(3\bar{q} + 4)$ is nonnegative.

A direct certificate now covers the entire $U^2 \leq V$ chart, so the secondary P, Q, C complements need no additional covering argument. In (D, W, T, p, q, A_0), F has multidegree $(11, 7, 3, 2, 2, 2)$ and 356 terms, while J has multidegree $(16, 8, 3, 4, 1, 3)$ and 1094 terms. Put

$$K = J - 4F, \qquad M = -K_{DW}\big|_{T=1,q=0,D=W=0},$$
$$J = 4F + M(D - W/2)^2 + E,$$
$$M = -16A_0p(2220A_0^2p^3 - 1190A_0p^3 - 1170A_0p + 161p + 456).$$

The complete tree has 95 leaves at depth at most 29: 49 use $-F \geq 0$, 35 use $K \geq 0$, 9 use $J \geq 0$, and 2 use $M, E \geq 0$. All integer slices and condition-zero-face Jacobian audits pass. Together with the opposite chart this proves $F \geq 0 \Rightarrow J \geq 0$ throughout both compact cubes.

It remains to orient the stationary root; $S(1) \geq 0$ must not be assumed without proof. The exact cleared polynomials satisfy

$$F(q) = F(0) + q\,C_qH(q),$$

where H is affine in q, C_q is independent of q, and

$$C_q = \begin{cases} (1 - T)(3U + 1)(13U^3W + 3U^2W + 12U + 2), & V \leq U^2, \\ (1 - T)(3DW + 1)(13D^3W + 3D^2 + 12DW + 2), & U^2 \leq V. \end{cases}$$

Both factors are strictly positive for $0 < T < 1$. Moreover $F(0)$ is a positive multiple of S_0. Suppose $S = 0$ at $0 < B_0 \leq 1$ and $S_1 \leq 0$. Then $S_0 = -B_0S_1 \geq 0$, so the strict slope implication at $q = 0$ gives $H(0) > 0$. If $S_1 = 0$, the displayed identity at the actual q gives $F(q) = F(0) \geq 0$ with $H(q) = 0$, a contradiction. If $S_1 < 0$, affine dependence gives a root of H between 0 and the actual q; there again $F = F(0) \geq 0$, contradicting strictness. Consequently $S_1 > 0$, and

$$S(1) = (1 - B_0)S_1 \geq 0, \qquad R = J/S_1 \geq 0, \qquad \mathcal{G}_4 = (n + 3)PR \geq 0.$$

Proposition 21 (Feasible positive interior at level four)**.** *For every finite $n \geq 4$, every feasible positive stationary point of* (90) *satisfying the reduced third-moment child bounds has nonnegative parent gap. The proof is an exact algebraic certificate; it has not yet been formalized in Lean.*

This closes the interior polynomial obligation. A claim for the entire uniform renewal cone still requires auditing the reduction of its general constrained boundary minima to the special slack faces discussed below. The identities and certificate generator are in `python/check_k4_uniform_interior.py`. The P checker is `python/check_k4_p_radial_bernstein_tree.cpp`; the shared streamed checker is `python/check_k4_radial_bernstein_tree.cpp`, with the complete midpoint trees stored in `python/certificates/`.

There is now an exact result at the fully sharp corner of the reduced cone. Put $a = rA$, $b = B$, $c = rC$, $d = D$, and write e, f, g, j for the aggregated return and terminal moments of orders four and five. The compact recurrence continues as

$$\begin{aligned} M_4 &= 4TM_3 + 6aM_2 + 4cM_1 + e + f, \\ M_5 &= 5TM_4 + 10aM_3 + 10cM_2 + 5eM_1 + g + j. \end{aligned}$$

The lower hierarchy and $T \geq ru$ imply the seven reduced inequalities

$$\begin{gathered} nTc \geq (n+1)a^2, \quad (n+1)ae \geq (n+2)c^2, \quad (n+2)cg \geq (n+3)e^2, \\ (n-1)b \geq nv^2, \qquad\qquad nvd \geq (n+1)b^2, \quad (n+1)bf \geq (n+2)d^2, \quad (n+2)dj \geq (n+3)f^2. \end{gathered}$$

Proposition 22 (Fully sharp $k = 4$ renewal corner)**.** *For every $n \geq 4$, if equality holds in all seven displayed reduced inequalities, then*

$$(n+3)M_3M_5 - (n+4)M_4^2 \geq 0.$$

The exact certificate splits at the Erlang scale $v = (n-1)T$. On the chart $T \geq v/(n-1)$, writing $T = (v+h)/(n-1)$ produces 608 nonnegative power coefficients. On the other chart, write $v = (n-1)T + h$. For $n \geq 8$, every coefficient is nonnegative except the middle term in

$$Aa^6 - BTha^5 + CT^2h^2a^4.$$

This quadratic is nonnegative because

$$4AC - B^2 = 4n^2(n-1)^{11}(n+1)^2(n+2)\big(3n^5 + 29n^4 + 111n^3 + 207n^2 + 182n + 132\big) > 0.$$

The orders $n = 5, 6, 7$ require respectively four, four, and two elementary AM–GM pairings. For $n = 4$, the regions $h \leq T$, $h \geq T$ with $a \leq Th$, and $h \geq T$ with $a \geq Th$ have nonnegative two-variable Bernstein coefficients. These checks are exact over $\mathbb{Q}$ in `python/check_k4_uniform_sharp_corner.py`.

The first full one-slack extension of this proposition is also available. Write

$$c = \frac{n+1}{n}\frac{a^2}{T} + s, \qquad s \geq 0,$$

and continue to impose equality in the other six reduced inequalities.

Proposition 23 (Return-third-moment slack face)**.** *For every $n \geq 4$, the parent gap remains nonnegative for every $s \geq 0$ under these assumptions.*

Here is a compact description of the exact certificate. After clearing a positive denominator, the gap is a quartic

$$p_0 + p_1 s + p_2 s^2 + p_3 s^3 + p_4 s^4.$$

The constant term is the fully sharp gap above, and

$$p_4 = T^4 n^4 (n-1)^6 (n+2) > 0.$$

After positive factors are removed, p_3 is a quadratic in a with positive leading coefficient and nonpositive discriminant for $n \geq 4$. The coefficient p_2 has one AM–GM block on each of the two scale charts. On the upper chart $T = (v+h)/(n-1)$, every coefficient of p_1 is nonnegative. On the lower chart $v = (n-1)T + h$, p_1 is coefficientwise nonnegative for $n = 5$; for every $n \geq 6$, its sole negative monomial is a multiple of

$$T^2 h a^5 s.$$

It is absorbed by one AM–GM pairing using only one eighth of the sharp a^8 coefficient and one half of a positive $T^4 h^2 a^2 s^2$ coefficient. The case $n = 4$ has two negative linear monomials; two pairings using one sixteenth of their neighboring coefficients leave the three earlier Bernstein charts nonnegative. The complete rational-arithmetic check is in `python/check_k4_uniform_c_slack.py`.

These two propositions still do not prove the uniform renewal lemma. Allowing slack in the terminal second- and third-moment inequalities (the inequalities defining b and d) produces two boundary faces and a possible feasible interior minimum of the top (e, f)-quadratic. The common equality boundary, the entire return-c-slack face, and the entire terminal-d-slack face are now settled in the stated sharp-face settings. The feasible positive interior minimum is also settled above; the reduction of every general constrained boundary minimum to these special faces still requires an audit.

The terminal-d-slack face reduces to a small four-variable positivity problem whose exact certificate is now complete. Keeping the other six bounds sharp, write

$$a = zT^2, \qquad d = \frac{n+1}{n}\frac{b^2}{v}(1+r), \qquad z, r \geq 0.$$

After homogeneity removes v, the variables are $n-4, T/v, z, r$. The only difficult locus found by adaptive Bernstein subdivision is

$$(n-1)T = v, \qquad z = r = 0.$$

If h denotes the signed deviation from this Erlang scale, the gap has positive first-order term

$$\frac{n^2(n+1)^2(n+2)(n+3)}{(n-1)^7}(z + 6r),$$

while the scale deviation begins at order h^2. This leads to the weighted blow-up $z = h^2 Z,\ r = h^2 R$. Three projective charts, according as h^2, z, or r is largest, remove the local singularity. Numerical Bernstein subdivision then certifies every compact chart, but unstructured exact subdivision accumulates at equality boundaries. Weighted blow-ups, rather than deeper rectangular subdivision, resolve these accumulations. The reproducible exploration is in `python/probe_k4_uniform_d_slack.py`; the completed exact charts are checked by `python/check_k4_uniform_d_slack.py`.

The two tail normalizations have been made explicit. In the T-dominant chart set $T = 1$, $v = (n-1)q$, $a = z$, and use the actual slack $D = s/T^3$. The entire region

$$0 \leq q \leq 1, \qquad 0 \leq D \leq 1$$

is now certified exactly for every $n \geq 4$. Near the return-only boundary the correct ratio is $R = D/q^3$. The chart $D = q^3R$ is coefficientwise nonnegative in both the bounded-z and reciprocal-z charts. In the complementary chart put $D = h^3$, $q = hQ$. Its corner has positive boundary orders h^5 and Q^4, so the two charts

$$(h, Q) = (t^4H, t^5), \qquad (h, Q) = (t^4, t^5R)$$

are polynomial. The first is again coefficientwise nonnegative. In the second, the degree drop at $n = \infty, R = 0$ is resolved by $x = 1/(n-3)$ and the two ordinary charts $R = xW$, $x = RW$. The only remaining projective corner there has coordinates $U = 1/z, W$, whose leading orders are U^4 and W^4; its two ordinary blow-up charts close after one midpoint split. All these assertions are exact integer power-to-Bernstein computations in `python/check_k4_uniform_d_slack.py`.

It remains on this T-normalized half to certify the large-slack tail $D \geq 1$. Its natural weighted normalization is $H = D^{-1/3}$, $Z = zH^2$, not the ordinary ratio z/D. Numerical Bernstein subdivision finds no negative value in either the bounded-Z or reciprocal-Z chart. The singular faces $H = 0$, $q = 0$, and $U = 1/Z = 0$ are coefficientwise nonnegative. At the remaining corner $H = Z = 0$, the boundary orders are H and Z^2. Both weighted charts

$$(H, Z) = (t^2A, t), \qquad (H, Z) = (t^2, tA)$$

have coefficientwise nonnegative faces at $t = 0$ and $q = 0$. The formerly missing $A = 0$ face of the second chart is the original face $z = 0$, and it can also be closed exactly. After removing the positive factor $(N+3)^8(N+4)^2$, where $N = n-4$, its gap is a quartic in D. Expanding instead in $E = D-1$, all five E-coefficients are nonnegative. The only corner not visible in their initial Bernstein arrays is $N = \infty, q = 0$; with $x = 1/(N+1)$, the two ordinary charts $x = qR$ and $q = xR$ certify every coefficient (only the linear E-coefficient needs a dyadic split, into three boxes). This gives a self-contained exact certificate for $z = 0, D \geq 1$ in `python/check_k4_uniform_d_slack_z0.py`.

Their common residual degree staircase at $n = \infty, t = 0$ is governed by $1/n \sim t^3$. Put $x = 1/(n-3)$ and use the two charts $x = r^3, t = rB$ and $t = r, x = r^3B$. All resulting powers of r are multiples of three. With $y = r^3$, the lower Newton monomials at the next corner are

$$q^7, \qquad yq^4, \qquad y^2q^2, \qquad y^3.$$

The two Newton slopes give three toric regions $y \leq q^3$, $q^3 \leq y \leq q^2$, and $y \geq q^2$. The corresponding monomial charts are now explicit. All of their auxiliary $B = 0$ faces are coefficientwise nonnegative; so are all $A = 0$ faces except the one already supplied by the $z = 0$ certificate above. These exact face checks are collected in `python/check_k4_uniform_d_slack_large_faces.py`. Thus the remaining work initially consisted of the interiors of twelve toric charts, with all accumulating boundary faces already proved. Their unsplit Bernstein arrays still contain negative coefficients, so a face-aware decomposition or controlled exact subdivision is required; direct rectangular subdivision produces an impractically large certificate.

One of those twelve charts is now completely closed. In the H-dominant, t-dominant chart with $y \leq q^3$, all negative Bernstein coefficients accumulate at $q = A = 0,\ R = B = 1$. Its next Newton terms are

$$A^4, \qquad A^3q, \qquad A^2q^2, \qquad Aq^3, \qquad q^6.$$

The polynomial has the manifestly positive factor

$$(1 + 2BR^3q^3)^8.$$

After removing it, the three regions $A \le q^3$, $q^3 \le A \le q$, and $q \le A$ have exact nonnegative Bernstein coefficients. The latter two would create large dense four-dimensional tensors, so the conversion is performed one B-slice at a time. The complete low-memory certificate is `python/check_k4_uniform_d_slack_large_ht_first.py`.

Two further toric interiors close after the common positive factor $(1+2w^6)^8$ is removed. The H-dominant, x-dominant chart with $y \ge q^2$ requires five terminal dyadic boxes and depth at most six; the corresponding Z-dominant, x-dominant chart requires nine boxes and depth at most ten. Their exact checkers are `python/check_k4_uniform_d_slack_large_hx_last.py` and `python/check_k4_uniform_d_slack_large_zx_last.py`. Thus three of the original twelve interiors are complete at this stage.

The Z-dominant, t-dominant chart with $y \le q^3$ is also complete. After removing $(1+2BR^3q^3)^8$, two dyadic splits in A certify the whole region $1/4 \le A \le 1$; see `python/check_k4_uniform_d_slack_large_zt_first_large_a.py`. At $A = 0$ the reduced polynomial factors as

$$q^2R^9B^{10}(1+3BR^3q^3)(1+4BR^3q^3)(1+5BR^3q^3)(1+6BR^3q^3)^2.$$

The nondominated exponent vectors in the order (A, q, R, B) are

$$(0,2,9,10),(1,1,6,8),(2,0,3,6),(4,1,3,5),$$
$$(5,0,0,3),(7,1,0,2),(9,2,0,1),(11,3,0,0).$$

The small-A tail is a genuine multiscale corner. Its first comparison is $A \sim qR^3B^2$, and the reason for this scale is especially transparent: the first three Newton monomials, including their signs, are exactly

$$q^2R^9B^{10} - 2AqR^6B^8 + A^2R^3B^6 = R^3B^6(A - qR^3B^2)^2.$$

The face factorization, Newton exponents, and leading square are checked exactly in `python/check_k4_uniform_d_slack_large_zt_first_newton.py`.

These scales lead to a short complete cover. Below the first surface, put $A = qR^3B^2C$ and $U = 1 - C$. The only residual equality corner has Newton orders

$$U^2, \qquad q^3U, \qquad q^4.$$

The three associated charts $U \le q^3$, $q^3 \le U \le q$, and $q \le U$ have exact nonnegative Bernstein certificates; only a depth-twelve dyadic refinement of the first chart is needed. This is checked in `python/check_k4_uniform_d_slack_large_zt_first_lower.py`.

For $qR^3B^2 \le A \le RB$, write $A = RBX$. The condition becomes $qR^2B \le X$, and four ordinary projective charts give direct streamed Bernstein certificates; see `python/check_k4_uniform_d_slack_large_zt_first_middle.py`. Finally, $RB \le A$ is covered by the two charts $R \le A$ and $A \le R, RB \le A$, checked in `python/check_k4_uniform_d_slack_large_zt_first_upper.py`. Hence four of the original twelve toric interiors are now complete and eight remain.

The adjacent H-dominant, t-dominant middle chart is likewise complete. Removing $(1+2BQ^6C^3)^8$ leaves the boundary factor

$$A^2B^6(A - B^2C)^2 \qquad (Q = 0).$$

On the side $A \le B^2C$, the substitution $A = B^2CX$, $U = 1 - X$ has only the transverse Newton orders U^2 and Q^6; the two charts at $U = Q^3$ have nonnegative exact Bernstein coefficients. Three ordinary projective charts cover $B^2C \le A$. The streamed certificate is `python/check_k4_uniform_d_slack_large_ht_middle.py`. Thus five of the twelve toric interiors are complete and seven remain.

The final H-dominant, t-dominant chart is complete as well. After removing $(1+2Bw^6)^8$, its boundary factor is

$$A^2B^6(AC^2-B^2)^2 \qquad (w=0).$$

Four monomial projective charts cover the two sides of $AC^2=B^2$. One chart requires a single dyadic split of its ratio variable; the five resulting streamed Bernstein checks are all nonnegative. See `python/check_k4_uniform_d_slack_large_ht_last.py`. Thus six of the twelve toric interiors are complete and six remain. The exploratory transformations are in `python/probe_k4_uniform_d_slack_large_toric.py`. In the Z-dominant, x-dominant chart with $y \le q^3$, the Newton boundary starts with

$$R^3(AB^3-qR^3)^2 + A^5B^{12}.$$

Five monomial charts give a complete exact certificate on the side $qR^3 \le AB^3$; see `python/check_k4_uniform_d_slack_large_zx_first_above.py`. On the opposite side put

$$B = R(1-U), \qquad A = q(1-V).$$

After the common nonnegative factor q^2R^9 is removed, its leading core is

$$(3U+V)^2 - 12(3U+V)q^3R^3 + 8q^4R^6.$$

Thus the transverse scale is $\max\{U,V\} \asymp q^2R^3$. Split into $V \le U$ and $U \le V$, and write $D = \max\{U,V\}$, with the smaller deficit equal to DZ. For $D \le q^2R^3$, put $D = q^2R^3W$. The resulting compact chart has a direct streamed Bernstein certificate on the three R-intervals

$$[0,1/2], \qquad [1/2,3/4], \qquad [3/4,1];$$

only the last needs one midpoint split in W.

For $q^2R^3 \le D$, five regular monomial charts give a complete cover:

$R \le D$	$R = DS,$
$R^2 \le D \le R$	$R = ST, \quad D = S^2T,$
$R^3 \le D \le R^2$	$R = TL, \quad D = T^3L^2,$
$qR^3 \le D \le R^3$	$D = R^3L, \quad q = LM,$
$q^2R^3 \le D \le qR^3$	$D = R^3M^2N, \quad q = MN.$

After division by the corresponding leading monomial, all ten charts (five for each deficit ordering) have nonnegative exact Bernstein coefficients. The four-dimensional conversions are streamed one final-axis slice at a time, and the five large charts are constructed and discarded sequentially. The certificate for this side is `python/check_k4_uniform_d_slack_large_zx_first_below.py`; the wrapper `python/check_k4_uniform_d_slack_large_zx_first.py` checks both sides. Hence the first Z-dominant, x-dominant interior is complete: seven of the original twelve toric interiors are now closed and five remain.

The corresponding first H-dominant, x-dominant interior is now complete as well. The same two-sided projective cover applies after canceling the smaller common monomials appropriate to the H-chart. On the side $qR^3 \le AB^3$, the five streamed polynomials have only 751 terms each. On the opposite side, the two compact charts and ten large-deficit charts have 15948 terms each; their maximal degrees are strictly smaller than those of the Z-dominant certificate. Every streamed Bernstein coefficient is nonnegative, with the same single midpoint split of W on the top compact R-interval. The complete low-memory checker is `python/check_k4_uniform_d_slack_large_hx_first.py`. Thus eight of the twelve toric interiors are complete and four remain.

The middle H-dominant, x-dominant interior is now complete too. After removing $(1+2Q^6C^3)^8$, every exponent of Q and B is a multiple of three. Thus the substitutions

$$P = Q^3, \qquad K = B^3$$

reduce the degrees 267 and 183 in the first attempted certificate to 89 and 61. At $P = 0$ the compressed polynomial has the exact face

$$A^2(AK - C)^2.$$

On $C \le AK$, put $C = AKX$, $U = 1 - X$; the two ordinary charts $U = PS$ and $P = US$ have nonnegative streamed Bernstein coefficients. The opposite side is covered without denominators by

$$A \le C: \quad A = CX, \qquad C \le A: \quad C = AY, \quad K = YX.$$

In the first chart one orders the deficits $1 - X$ and $1 - K$, then splits their maximum against P. In the second chart the first residual scale is again $U \sim P$. Its only further boundary accumulation has face S^2 at $Y = 0$, hence the weighted split $Y \sim S^2$. The remaining Newton boundary is

$$R^2, \qquad T^5R, \qquad T^6,$$

so a final split at $R = T^3$, followed by a square compression of the radial variable, regularizes the corner. Three rational intervals and one shallow terminal subdivision close the two far boxes. The complete low-memory certificate is `python/check_k4_uniform_d_slack_large_hx_middle.py`. Consequently nine of the twelve toric interiors are complete. The three remaining ones are the middle Z-x chart and the middle and final Z-t charts.

The middle Z-dominant, x-dominant interior is now complete as well. The same cube compression gives a (751)-term polynomial of degree $(38, 53, 13, 8)$ in (P, C, A, K), but its face is now

$$(AK - C)^2.$$

The side $C \le AK$ is again covered by $C = AKX$, $U = 1-X$, and the two charts $U = PS$, $P = US$. On the opposite side, the charts $A = CX$ and $C = AY, K = YX$ give a polynomial cover of $A \le C$ and $C \le A$, respectively. In the first chart, order (1-X) and (1-K) and compare their maximum with P. Two of the four resulting charts are direct. The other two have the Newton boundary

$$C^4, \qquad C^3W, \qquad W^2,$$

which is resolved by

$$W = C^3T, \qquad (C, W) = (LT, LT^3), \qquad C = WT.$$

The low chart uses four rational T-intervals, the middle chart two dyadic intervals, and the high chart is direct.

For $C \le A$, the successive scales are $U \sim P$, $Y \sim S^2$, and $R \sim T^3$, exactly as in the H-dominant calculation. On the low R-chart, $W \ge 1/2$ is direct after three rational intervals. After rescaling the remaining half to a unit box, its only unresolved face has Newton monomials

$$A^4, \qquad A^3W, \qquad W^2,$$

and the three charts

$$W = A^3T, \qquad (A, W) = (LT, LT^3), \qquad A = WT$$

all have nonnegative streamed Bernstein coefficients. The opposite $T^3 \le R$ chart closes on two dyadic radial intervals. The complete low-memory checker, with independently selectable sections, is `python/check_k4_uniform_d_slack_large_zx_middle.py`. Consequently ten of the twelve toric interiors are complete; only the middle and final Z-t charts remain.

The final Z-dominant, t-dominant interior is now complete. After removing $(1 + 2Bw^6)^8$, its face at $w = 0$ is

$$B^6(AC^2 - B^2)^2.$$

Four ordinary projective regions cover the two sides of this equality. The only additional accumulation occurs in the $B \le C$ region. After the common factor X^{10} is removed, its equality deficit has the weighted scale $1 - Y \asymp w^3$. The high chart $w^3 \le 1 - Y$ is direct. In the low chart the Newton boundary is

$$S^2, \qquad 8X^2, \qquad -12Cw^3XS, \qquad C^2w^6.$$

Comparing S, X, and w^3 gives a three-way weighted fan. The S-dominant chart is direct, the X-dominant chart closes by a depth-twelve streamed subdivision, and the w^3-dominant chart reduces to the positive face $V^2 + 8U^2 + C^2$. An ordinary three-way fan comparing U, V, C resolves that face; only the U- and C-dominant charts need depth-twelve subdivision. The ten independently selectable checks are in `python/check_k4_uniform_d_slack_large_zt_last.py`. Consequently eleven of the twelve toric interiors are complete; only the middle Z-t interior remains.

The middle Z-dominant, t-dominant interior is complete as well. After removing $(1+2BQ^6C^3)^8$, its face at $Q = 0$ is

$$B^6(A - B^2C)^2.$$

On $A \le B^2C$, put $A = B^2CX$ and $U = 1 - X$. The far chart $Q^3 \le U$ is direct. In the near chart $U \le Q^3$, the remaining Newton boundary is

$$C^4, \qquad C^3S, \qquad S^2.$$

The associated low, middle, and high fan charts cover $S \le C^3$, $C^3 \le S \le C$, and $C \le S$. The middle chart closes on the two dyadic halves of its radial variable, and the high chart is direct. In the low chart, after the cube compression $P = Q^3$, the secondary Newton support is

$$B^2, \qquad CP^2, \qquad C^2.$$

Eight ordinary projective charts compare these three scales. Their four-dimensional Bernstein conversions are streamed one final-axis slice at a time; only terminal slices require depth-eighteen subdivision. The opposite side $B^2C \le A$ is covered by three ordinary projective charts, all direct after their common positive monomial factors are removed. The complete low-memory certificate, with independently selectable sections and fixed degree-and-support assertions, is `python/check_k4_uniform_d_slack_large_zt_middle.py`.

Consequently all twelve toric interiors, together with their previously certified boundary faces, are closed. This completes the T-normalized large-slack tail of the terminal-d-slack face.

In the v-dominant chart set $v = 1$, $T = q/(n-1)$, $a = zT^2$, retain the relative slack r, and divide the cleared numerator by its common factor q^4. A further ordinary projective split according as 1, z, or the slack is largest removes the artificial degree drop at infinity. The central and slack-dominant charts are direct. The return-dominant chart has only the corner $N = q = U = V = 0$. Its Newton facets in (q, U, V) have primitive normals

$$(1, 2, 1), \qquad (2, 3, 3), \qquad (4, 5, 5).$$

After ordering U, V, the side $V \le U$ is covered by the radial scales $R \asymp q^{3/2}$ and $R \asymp q^{5/4}$, while the side $U \le V$ has a homogeneous degree-eight three-way fan. The infinity faces of these charts are resolved by the successive Newton normals $(1,1),(2,5)$, then by a homogeneous degree-three fan, the $(2,3)$ face, and finally the $(1,4,3)$ face.

The last accumulation occurs in a low-t, A-dominant ratio chart. Its (A,R)-Newton boundary is

$$(0,24), \qquad (1,18), \qquad (2,0),$$

with respective leading coefficients $1, -2Xt^4, 1+X$. The middle point lies strictly above the facet of normal $(12,1)$. Hence the two weighted charts $A = R^{12}B$ and $A = D^{12}, R = DB$ have positive leading faces $1+B^2(1+X)$ and $B^{24}+1+X$, respectively, and their exact Bernstein certificates close after four and two subdivisions. The analogous residual ρ-chart has boundary $(0,4),(1,3),(2,0)$ and is closed by the two charts associated with the normal $(2,1)$. Replacing the recurring powers R^6, ρ^6, t^2 by new unit-interval variables keeps the largest dense certificate small. Fixed degree-and-support assertions and the complete exact checks are in `python/check_k4_uniform_d_slack_v_tail.py`.

Together with the T-normalized certificates above, this completes the entire terminal-d-slack boundary face under its six sharp bounds. The feasible positive interior minimum is now certified above. Before claiming the uniform $k = 4$ renewal lemma, the general boundary reduction must still be justified.

The exact constrained-boundary audit keeps the lower moments free. Normalize $M_1 = 1$, put $\alpha = (n+3)/(n+2)$, $\beta = (n+2)/(n+1)$, and write

$$y_0 = \beta c/a, \qquad f_0 = \beta d/b, \qquad \delta_0 = 5/(2\alpha).$$

For $Q = \mathcal{G}_4$ in the ratio coordinates (y, f),

$$\frac{Q_y}{c} - \frac{Q_f}{d} = 2(n+3)P\alpha(y - f + \delta_0).$$

Thus a minimum on $y = y_0$ with $f > f_0$ must have $Q_f = 0$ and $f - y_0 \le \delta_0$. Its free-coordinate root is

$$f_e = \frac{2(n+4)(L + cy_0) - 5(n+3)PT}{2\{(n+3)P\alpha - (n+4)d\}}.$$

Likewise a minimum on $f = f_0$ with $y > y_0$ has $Q_y = 0$ and $f_0 - y \ge \delta_0$, with root

$$y_e = \frac{2(n+4)(L + df_0) - 5(n+3)P(T+1)}{2\{(n+3)P\alpha - (n+4)c\}}.$$

Both denominators are strictly positive. Completing the two squares is exact; the cleared edge-root values have respectively 967 and 1078 terms in (n, T, a, b, c, d). Their conditional nonnegativity under root feasibility and the displayed gradient signs remains open, as does the general corner $y = y_0, f = f_0$ with simultaneous lower-moment slacks. These identities and support assertions are reproducible in `python/check_k4_uniform_top_boundaries.py`; they are reductions, not positivity certificates for the missing faces.

An additive-cone shortcut does not close this gap. If the moment triple is split into its return-only part X and terminal-response part Y, the return-only part satisfies the desired log-convexity inequality, but Y need not. Already at $n = 4, T = 1, v = 2, a = s = 0$, its corresponding gap is $-819136/243$. Thus closure of the log-convexity cone under addition cannot replace the remaining cross-term analysis.

Monotonicity or convexity in the terminal slack also fails. For the cleared T-normalized numerator, at $n = 4, q = 1, z = 3, D = 1$ the first D-derivative is -1804714429536. At $n = 6, q = 1, z = 11, D = 122$, the second derivative is -490528950000000. Consequently the certified face $D = 1$ cannot simply be propagated through the large-slack tail by either argument.

One should not try to repeat the next convexity step from the $k = 3$ proof. After setting

$$E = \frac{n+2}{n+1}\frac{C^2}{A}, \qquad F = \frac{n+2}{n+1}\frac{D^2}{B},$$

the reduced gap is not convex in C, D. For example, at

$$n = 4, \quad x = r = u = v = 1, \quad A = \frac{4}{3}, \quad B = 2, \quad C = \frac{20}{9}, \quad D = 5,$$

its Hessian is

$$\begin{pmatrix} 558194/15 & -1975/2 \\ -1975/2 & -155882/15 \end{pmatrix}, \qquad \det = -\frac{348927229057}{900}.$$

The gap itself is nevertheless positive there. The interior stationary lemma above is now certified, but nested convexity cannot supply the missing general boundary reduction. All identities and this exact witness are checked by `python/probe_k4_uniform_renewal.py`.

8 The acyclic case

Conjecture 2 has a short proof when the directed graph of the off-diagonal entries of A is acyclic. Expanding an entry of e^{tA} by directed paths, a path with ℓ off-diagonal edges contributes

$$t^{\ell} \int_{\Delta_\ell} \exp\left(t \sum_{j=0}^{\ell} x_j a_{i_j i_j} \right) d\mu(x),$$

up to a positive constant. Since the path is simple, $\ell \le s-1$. After division by t^{s-1}, every integrand is of the form

$$t^{-(s-1-\ell)} e^{ct},$$

which is log-convex. Positive integrals and positive sums of log-convex functions are log-convex (by Hölder's inequality). This recovers the triangular result and shows precisely where cycles enter: a cyclic walk may have arbitrarily many off-diagonal edges, so the exponent $s - 1 - \ell$ is no longer nonnegative path by path.

9 Evidence and completion in dimension three

The curvature in (5) was tested numerically on random dense and sparse nonnegative matrices through size 6, including extreme rate ratios, deterministic initial and terminal vectors, directed cycles, and direct optimization of the curvature. No negative value was found; serial acyclic chains approach equality. This is evidence only, not a proof.

In dimension three, Proposition 14 makes a search over general matrices unnecessary. Direct optimization of the three canonical curvatures over widely separated rates and feedback parameters likewise found no negative value. Taylor-coefficient experiments reveal a more precise obstruction: after the shift leading to (14), the second and third start states have many manifestly nonnegative coefficient groups, whereas the first start has negative monomials beginning in feedback degree two.

The full coefficient and the pointwise curvature remain nonnegative in all tests. Thus a proof must group unequal-rate terms (typically into expressions involving $(x_i - x_j)^2$); positivity monomial by monomial is already false in this reduced chart.

Proposition 20 and the real-root divided-difference argument settle the middle start for every spectrum; convolution then settles the third start. Proposition 19 now settles the first start throughout the complex-spectrum region as well. The degree-60 certificate following (20) also settles the real-spectrum first start through Perron-normalized time 10. Proposition 16, the coefficient comparison following (41), and the exact low-degree block (42) settle its unbounded tail.

Corollary 24 (Complete three-state case)**.** *Conjecture 2 holds for every* 3×3 *nonnegative matrix. Consequently, through Proposition 1, the full moment hierarchy holds for every three-state phase-type law.*

Formalization status. The Lean development now proves normalized complete-symmetric log-convexity for every index. The exact block covers $1 \le n \le 5$, and the transition argument above covers $n \ge 6$. For the last transition-point estimate, the formal proof uses the simpler majorant

$$b_n \le \frac{n-1}{n+1} \qquad (n \ge 7),$$

deduced from $(1 + 1/m)^m \le e < 3$; the case $n = 6$ is an exact tenth-power comparison. This avoids formalizing the auxiliary real-variable monotonicity argument while proving the same bound. The combined Lean theorem is `normalizedComplete_logConvex_all`. The subsequent weighted Chebyshev reduction has also been formalized: all dependence on u, v is discharged. The parameter-free sum in (41) is now evaluated in Lean by expanding its kernel in the first five falling-factorial polynomials and using the binomial falling-factorial moments. Consequently the formal far-tail coefficient theorem has no unevaluated finite-sum hypothesis.

The remaining finite work has since been shortened further. After restoring the constant term $h_1 = 1 + w$, the full corrected-curvature coefficients are nonnegative directly: degrees $N = 6, 7, 8$ have short algebraic proofs, degrees $N = 9, \dots, 14$ use six independent two-variable Bernstein certificates, and the general comparison proves every $N \ge 15$. The certificates are isolated in separate modules so that they can be checked with bounded peak memory. Their assembly is `fullCurvatureCoefficient_nonneg_of_root`.

Absolute convergence and two termwise differentiations are also formalized. The full coefficient series is identified with

$$t^2\{h_1 h_1'' - (h_1')^2\} + 2h_1^2,$$

and summed term by term. Thus `firstStartCurvature_nonneg_of_root_all_time`, and equivalently `normalizedFirstStart_correctedCurvature3_nonneg`, proves the real-spectrum first start for every $t \ge 0$; the former restriction $t \ge 10$ and the degree-60 compact-time certificate are no longer needed.

The normalization back toward the canonical cubic is now partly formalized as well. The file `K3CyclicRealCanonicalBridge.lean` proves amplitude, time-scale, and exponential-shift transport, including a version stated with actual first and second derivatives. From Vieta's identities and the canonical sign conditions it proves that ordered real roots are nonnegative and that their ratios $u = \lambda_1/\lambda_3$, $v = \lambda_2/\lambda_3$ satisfy $\sqrt{u} + \sqrt{v} \le 1$. Consequently `canonicalOrderedRealRoots_firstStart_nonneg` is the checked pointwise first-start inequality for every ordered real-root canonical cubic. The remaining matrix-exponential identification is now formalized as well. The file `K3CyclicCanonicalMatrixBridge.lean` proves the canonical matrix cubic identity, differentiates

the actual first coordinate of the matrix exponential three times, and invokes uniqueness for the resulting scalar cubic ODE. Its theorem `canonicalFirst_eq_orderedRealRootSeries` identifies that coordinate with the normalized Taylor series, while `canonicalFirst_orderedRealRoots_correctedCurvature3_nonneg` states the real-spectrum certificate directly for the actual first coordinate and its actual derivatives.

The complex first start is now complete in Lean. The lightweight identity module `K3CyclicComplexFirst.lean` contains the oscillator energy, Wronskian factorization, and reduction to (49); the certificate and analytic-tail modules cover the compact and large-x regions. Their assembly in `K3CyclicComplexFirstFull.lean` is `firstH_correctedCurvature3_nonneg`. The ODE-uniqueness bridge `K3CyclicComplexCanonicalFirstBridge.lean` now identifies this normalized function with the actual first coordinate of the canonical matrix exponential. The theorem `canonicalFirst_complexPair_correctedCurvature3_nonneg_of_vieta` in `K3CyclicComplexCanonicalFirstCurvature.lean` transports the certificate to that actual coordinate; its compact-domain hypotheses are derived internally from the canonical sign conditions and the complex Vieta identities.

The coefficient inequality behind the complex middle start is also now formalized. The exact conjugate pairing is proved in `K3CyclicComplexMiddlePairingReal.lean`. Separate low-memory Taylor modules prove (62), including the sharp third- derivative estimate for $(1-it)^2(1+it)^{N-2}$. The finite Bernstein certificate covers $4 \le N \le 24$, while `K3CyclicComplexMiddleTailCoefficient.lean` combines the analytic tail, the numerical domination for $N \ge 25$, and the degenerate $\omega = 0$ boundary. Its theorem `scalarCurvatureCoefficient_nonneg_all` proves every middle curvature coefficient is nonnegative throughout the compact chart. The EGF and Cauchy- product modules `K3CyclicComplexMiddleSeries.lean` and `K3CyclicComplexMiddleCurvatureSeries.lean` now sum those coefficients and prove `middleRecurrenceCurvature_nonneg`, the analytic-series form of (57). That identification is now complete: `K3CyclicComplexCanonicalMiddleBridge.lean` differentiates the shifted and scaled EGF, matches its cubic ODE and initial data to the actual middle coordinate, and transports the stronger curvature inequality. The separate low-memory module `K3CyclicComplexCanonicalMiddleChart.lean` proves the two square identities giving the exact interval (58), constructs the compact chart variables from the complex Vieta data, and concludes with `canonicalMiddle_complexPair_middleCurvature3_nonneg`. Finally, `K3CyclicCanonicalThird.lean` formalizes the actual third coordinate and the driven identities $h_3' = x_3 h_2 + b h_3$ and $h_3'' = x_3 h_2' + b h_3'$. The companion module `K3CyclicCanonicalThirdIntegral.lean` proves the exact variation-of- constants formula used above. The remaining formal work is the general convolution log-convexity lemma carrying the stronger middle inequality to the third coordinate, followed by the final canonical-to-general matrix reduction.

The paper *Log-Convex Matrix Functions* by Aujla, Rawla, and Vasudeva uses "matrix log-convexity" in a different sense: it studies the functional calculus $A \mapsto f(A)$ for positive-definite Hermitian matrices. Its main criteria do not apply to scalar coefficients $u^T e^{tA} v$ with a nonsymmetric nonnegative matrix A. Its closure of positive sums of log-convex functions agrees with the elementary closure used in the acyclic argument above, but does not prove Conjecture 2.

The paper by Butt, Pečarić, and Rehman on exponential convexity of Petrović-type functionals is also about a different class of convexity preservers. It contains no renewal, phase-type, or nonsymmetric semigroup result that applies to (10).